STA 6746
Multivariate Statistical Analysis
Lecture Notes

Shangyuan Ye

1 Introduction

Outline:

- Multivariate data
- Characteristics of multivariate data
- Multivariate methods

1.1 Multivariate data

- *Multivariate data* arise when multiple *response* variables are observed on each experimental unit.
  - Experimental unit: smallest entity to which a treatment or condition can be independently applied and from which a response is measured.
  - Variable: any characteristic or property of an experimental unit that can take on different values across units.
  - Response variable: the particular variable of primary interest whose variation you wish to explain or predict. In an experiment, the response is what you measure after applying treatments or conditions to experimental units.

Exercise: identify response/explanatory variables and experimental units for the examples below, and determine if the data are *multivariate* or *univariate*?

1. Several exam scores are recorded for each student in a class to gauge student performance.
2. Length, width, and volume of several skulls at 2 different archaeology sites are recorded. Scientists wonder if the skulls are from the same species.

3. Income, education level, and current debt are recorded for each loan applicant.
4. Blood pressure, daily caloric intake, ethnicity, and BMI were measured on patients in an effort to identify risk factors associated with high blood pressure.

1.2 Characteristics of multivariate data

- Multivariate methods are designed for the analysis of data with a certain *dependency structure*.
    1. Experimental units are sampled independently. What does this mean in the previous examples?
    2. Variables measured on each experimental unit are not independent. What does this mean in the previous examples?

1.3 Multivariate methods

We break multivariate methods down into *three scenarios*:

1. A single sample of experimental units from one population is taken.
2. Two samples of experimental units are taken; each sample is from a different population.
3. Multiple samples of experimental units are taken from multiple populations.

Scenario 1a: For a single sample, there are two sets of variables

- *Cannonical correlation analysis* is used to describe the correlation between two groups of variables.

- *Multivariate regression* is used to simultaneously model several dependent variables on several independent variables.

Example 1.1. *Several variables regarding a father's parental behavior and several other variables regarding his daughter's attitude could be measured.*

Scenario 1b: For a single sample, no groupings

- Perform hypothesis tests regarding the mean vector or covariance matrix of the data. Hotelling's T^2 test is useful here.
- Perform *Principle Component Analysis* and singular value decomposition. The goal of PCA is to describe the total variability of the data as much as possible with a few of its *principle components*. Principle components are the fundamental mathematical tool in many multivariate methods. For example, PCA techniques are used in the estimation of factors in a *factor analysis*, asses multivariate normality, and identify outliers.
- Perform *Factor Analysis*. Is there an underlying independent *factor(s)* or *unobserved latent variable(s)* governing the behavior of the dependent observed variables?

Example 1.2. *The IRS measures net income, tax deductions, claimed dependents, taxable income, tax paid, ... and would like to identify anomalies for potential audit.*

Example 1.3. *Spearmen published a paper in 1905 suggested that exams really measured the general intelligence of a student, plus the intelligence specific to the subject.*

Scenario 2: Two samples

- Are the two populations different? The answer to this question could determine how we proceed with an analysis. This involves a generalization of the usual two-sample Student's t test.
- *Discriminant Analysis*. The goal is to define a function that best "separates" experimental units according to the population from which they were drawn. In this step, it is known which population each unit came from.

- *Classification Analysis*. Now the population from which the unit was taken from is unknown. We seek to use a discriminant function to assign a unit to a population.

Example 1.4. *We may observe risk variables for home loan applications, such as current debt, percentage of income dedicated to mortgage payment, etc. We also observe whether or not a loan applicant eventually defaulted on his/her mortgage.*

Scenario 3: Multiple samples

- Each of the three types of methods for two samples can be generalized to handle k samples
- Are all populations the same? A *multivariate analysis of variance* can be used.
- *Multiple discriminant analysis* can be used to determine how k populations are different in terms of a collection of variables. For example, which variables contribute the most towards the discrimination between 2 or more of the populations?
- Discriminant functions for these k populations can be used to *classify* an experimental unit as belonging to 1 of k groups.

Comments:

- We will frequently be using a multivariate normal model for methods. This means that data should be continuous, or at least nearly continuous in nature.
- Most methods require lengthy calculations, and many different computing packages are available. Be aware that you may get different results from different packages, even though *you think* you are using the same method to analyze the data. The likely explanation is that, without knowing it, the methods you are applying differ in a subtle way. There are many subtle/minor but important technical details to most multivariate methods that can significantly alter your analysis. You should carefully read the manual and fully understand the method before you use any software.

2 Matrix algebra

Outline:

- Notations and definitions
- Basic operations on matrices
- Rank
- Inverses
- Quadratic forms
- Determinant and trace
- Orthogonality
- Eigenvalues and eigenvectors
- Singular value decomposition

2.1 Notations and definitions

- An $n \times p$ **matrix** is a rectangular array of numbers

$$\boldsymbol{A} = \boldsymbol{A}_{n\times p} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1p} \\ a_{21} & a_{22} & \dots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{2n} & \dots & a_{np} \end{pmatrix}$$

- Alternative notation is $\boldsymbol{A} = (a_{ij})$.
- A **vector** is a matrix with either 1 row or 1 column. A p-dimensional column vector is

$$\boldsymbol{x} = \boldsymbol{x}_p = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{pmatrix}$$

- A row vector is written $\boldsymbol{x}' = \boldsymbol{x}'_p = (x_1, x_2, ..., x_p) = \begin{pmatrix} x_1 & x_2 & \dots & x_p \end{pmatrix}$.
- A **scalar** is a 1×1 matrix, and could be used to represent a single observation.

- Two matrices $\boldsymbol{A} = (a_{ij})$ and $\boldsymbol{B} = (b_{ij})$ are **equal** if all of their elements are equal, i.e. $a_{ij} = b_{ij}$ for every i, j.
- Are the following two matrices equal?

$$\boldsymbol{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \boldsymbol{B} = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

- The **transpose** of an $n \times p$ matrix is a $p \times n$ matrix computed by interchanging rows and columns. More precisely, the transpose of $\boldsymbol{A}$ is $\boldsymbol{A}' = (a_{ij})' = (a_{ji})$. Notice that $\boldsymbol{A}' = \boldsymbol{B}$.
- A matrix $\boldsymbol{A}$ is **symmetric** if $\boldsymbol{A} = \boldsymbol{A}'$ or equivalently $(a_{ji}) = (a_{ij})$. A covariance matrix is an example of a commonly encountered symmetric matrix in statistics.
- A $p \times p$ **square matrix** $\boldsymbol{A} = \boldsymbol{A}_p$ is a matrix with the same number of rows and columns.
- A symmetric matrix must be a square matrix.

- The **diagonal** of a square matrix $\boldsymbol{A}_p$, written $diag(\boldsymbol{A}_p)$, is the original matrix with off-diagonal elements set to 0.
- A generic diagonal matrix is denoted by $\boldsymbol{D} = diag(d_{11}, d_{22}, ..., d_{pp})$ or $\boldsymbol{D} = diag(d_1, d_2, ..., d_p)$.
- The $p \times p$ identity matrix $\boldsymbol{I} = \boldsymbol{I}_p$ is a diagonal matrix of 1's.
- A vector of 1's of length n is denoted by $\boldsymbol{j} = \boldsymbol{j}_n$ and an $n \times n$ square matrix of 1's is denoted by $\boldsymbol{J} = \boldsymbol{J}_n$.
- A vector of 0's is denoted by $\mathbf{0}$ and a matrix of 0's is $\boldsymbol{O}$.

- Compare $diag(1,1,1)$, $\boldsymbol{I}_3$, and $diag(\boldsymbol{J}_3)$.

Code

```
> A<-matrix(c(1,2,3,4,5,6), nrow =3, byrow = T )
> A
     [,1] [,2]
[1,]    1    2
[2,]    3    4
[3,]    5    6
>B<-matrix(1:6, nrow = 2)
> B
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
> A==B
Error in A == B : non-conformable arrays
> t(B)
     [,1] [,2]
[1,]    1    2
[2,]    3    4
[3,]    5    6
> A==t(B)
     [,1] [,2]
[1,] TRUE TRUE
[2,] TRUE TRUE
[3,] TRUE TRUE
> I = diag(c(1,1,1))
> I
     [,1] [,2] [,3]
[1,]    1    0    0
[2,]    0    1    0
[3,]    0    0    1
> J = matrix(1, nrow = 3, ncol = 3)
> J
     [,1] [,2] [,3]
[1,]    1    1    1
[2,]    1    1    1
[3,]    1    1    1
> diag(J)
[1] 1 1 1
```

2.2 Basic operations on matrices

- The **addition** operation $\boldsymbol{C} = \boldsymbol{A} + \boldsymbol{B}$ is defined by $(c_{ij}) = (a_{ij} + b_{ij})$. It is only defined for matrices with equal dimensions.

- For c a scalar and $\boldsymbol{A}$ a matrix, define $c\boldsymbol{A} = \boldsymbol{A}c = (ca_{ij})$.

- To take the product $\boldsymbol{AB}$, $\boldsymbol{A}$ and $\boldsymbol{B}$ must be **conformable**. They are conformable if the number of columns of $\boldsymbol{A}$ is equal to the number of rows of $\boldsymbol{B}$. The (ij)th element of $\boldsymbol{C} = \boldsymbol{AB}$ is defined $c_{ij} = \sum_k a_{ik}b_{kj}$.

- Example: Define

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix} \text{ and } \boldsymbol{B} = \begin{pmatrix} -1 & 2 \\ 0 & 0 \\ 1 & 3 \end{pmatrix}.$$

- Compute $\boldsymbol{AB}$.

- Is $\boldsymbol{AA}$ defined?

- Compute $(\boldsymbol{AB})^2 = (\boldsymbol{AB})(\boldsymbol{AB})$.

- Compute $(\boldsymbol{AB})'(\boldsymbol{AB})$.

- Notice that multiplying a 2×3 matrix by a 3×2 matrix yielded a 2×2 matrix. In general

$$\boldsymbol{A}_{m\times p}\boldsymbol{B}_{p\times n} = \boldsymbol{C}_{m\times n}.$$

- The **Commutative** property rarely holds: $\boldsymbol{AB} \neq \boldsymbol{BA}$. In fact, $\boldsymbol{BA}$ may not even be defined. Consider $\boldsymbol{A} = (1, 2)$ and $\boldsymbol{B} = (0, 1)'$.

- The **Distributive** property holds: $\boldsymbol{A}(\boldsymbol{B} + \boldsymbol{C}) = \boldsymbol{AB} + \boldsymbol{AC}$. Consider $\boldsymbol{A} = (1, 2)$, $\boldsymbol{B} = (0, 1)'$, and $\boldsymbol{C} = (0, -1)'$

- The **Associative** property holds: $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC})$. Consider $\boldsymbol{A} = (1, 2)$, $\boldsymbol{B} = (0, 1)'$, and $\boldsymbol{C} = (0, -1)$

- The transpose of a product is the product of the transposes in reverse order: $(\boldsymbol{AB})' = \boldsymbol{B}'\boldsymbol{A}'$. Consider $\boldsymbol{A} = (1, 2)$ and $\boldsymbol{B} = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}$.
- The transpose of $\boldsymbol{A} + \boldsymbol{B}$ is the transpose of $\boldsymbol{A}$ plus the transpose of $\boldsymbol{B}$: $(\boldsymbol{A} + \boldsymbol{B})' = \boldsymbol{A}' + \boldsymbol{B}'$. Consider $\boldsymbol{A} = (1, 2, 3)$ and $\boldsymbol{B} = \boldsymbol{j}$.

Some important examples:

1. $\boldsymbol{j}'\boldsymbol{j} = n$.
2. $\boldsymbol{j}\boldsymbol{j}' = \boldsymbol{J}$.
3. Summing over experimental units for each variable can be done $\boldsymbol{j}'\boldsymbol{A}_{n\times p}$ or $\boldsymbol{A}'_{n\times p}\boldsymbol{j}$.
4. $\boldsymbol{AI} = \boldsymbol{A}$ and $\boldsymbol{IA} = \boldsymbol{A}$

Code

```
> A = matrix(c(1,2,3),nrow=3)
> j = matrix(1, 3, 1)
> j
     [,1]
[1,]    1
[2,]    1
[3,]    1
> j+j
     [,1]
[1,]    2
[2,]    2
[3,]    2
> 2*j
     [,1]
[1,]    2
[2,]    2
[3,]    2
> t(j)%*%j
     [,1]
```

```
[1,]    3
> j%*%t(j)
     [,1] [,2] [,3]
[1,]    1    1    1
[2,]    1    1    1
[3,]    1    1    1
> t(j)%*%A
     [,1]
[1,]    6
```

2.3 Rank

- Rank allows us to determine the "dimensionality" of a matrix, i.e. the number of linearly independent rows/columns of a matrix.
- Vectors $\boldsymbol{a}_1, \boldsymbol{a}_2, ..., \boldsymbol{a}_p$ are **linearly dependent** if there exists $c_1, c_2, ..., c_p$ (not all = 0) such that $c_1\boldsymbol{a}_1 + c_2\boldsymbol{a}_2 + \ldots + c_p\boldsymbol{a}_p = \mathbf{0}$ or equivalently if at least one of the vectors can be written as some linear combination of the remaining vectors. Otherwise they are **linearly independent**. Can you come up with two linearly independent vectors and two linearly dependent vectors?
- The **rank** of a matrix $\boldsymbol{A}$ is denoted $rank(\boldsymbol{A})$ and defined as the number of linearly independent rows or columns of $\boldsymbol{A}$. Consider $\boldsymbol{A} = \begin{pmatrix} 1 & -2 & 3 \\ 5 & 2 & 4 \end{pmatrix}$. Note that row 1 and row 2 are linearly independent, so the rank of $\boldsymbol{A}$ is 2. We should therefore be able to find, c_1, c_2, and c_3 such that

$$c_1 \begin{pmatrix} 1 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 2 \end{pmatrix} + c_3 \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

  It turns out that $c_1 = 14$, $c_2 = -11$ and $c_3 = -12$.
- What is $rank(\boldsymbol{I}_p)$ and $rank(\boldsymbol{J}_p)$?
- $rank(\boldsymbol{A}_{m\times n}) \leq min(m, n)$

- A matrix $\boldsymbol{A}_{m\times n}$ is of **full rank** if $rank(\boldsymbol{A}_{m\times n}) = \min(m,n)$. Is $\boldsymbol{I}$ of full rank? What about $\boldsymbol{J}$?
- $rank(\boldsymbol{A}') = rank(\boldsymbol{A})$
- $rank(\boldsymbol{AB}) \leq min\left\{rank(\boldsymbol{A}), rank(\boldsymbol{B})\right\}$

2.4 Inverse

- If $\boldsymbol{A}$ is a square matrix of full rank, it is said to be **nonsingular**, and its **unique inverse** $\boldsymbol{A}^{-1}$ satisfies

$$\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}.$$

- Is the inverse of

$$\boldsymbol{A} = \begin{pmatrix} 3 & 4 \\ 2 & 6 \end{pmatrix}$$

equal to

$$\boldsymbol{A}^{-1} = \begin{pmatrix} .6 & -.4 \\ -.2 & .3 \end{pmatrix}?$$

- If $\boldsymbol{A}$ is **nonsingular**, then $\boldsymbol{AB} = \boldsymbol{AC}$ implies $\boldsymbol{B} = \boldsymbol{C}$. That is, when solving, we can cancel out the $\boldsymbol{A}$. Verify this by multiplying both sides of the equation by $\boldsymbol{A}^{-1}$.
- If $\boldsymbol{A}$ is **singular**, then $\boldsymbol{AB} = \boldsymbol{CB}$ does not imply $\boldsymbol{A} = \boldsymbol{C}$. You can verify this by noticing that $\boldsymbol{AB} = \boldsymbol{CB}$ below, even though $\boldsymbol{A} \neq \boldsymbol{C}$.

$$\boldsymbol{A} = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \end{pmatrix}, \boldsymbol{B} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \boldsymbol{C} = \begin{pmatrix} 2 & 1 & 1 \\ 5 & -6 & -4 \end{pmatrix}$$

Code

```
> A<-matrix(c(3,2,4,6),2,2)
> A
```

```
     [,1] [,2]
[1,]    3    4
[2,]    2    6
> solve(A)
     [,1] [,2]
[1,]  0.6 -0.4
[2,] -0.2  0.3
> A%*%solve(A)
     [,1] [,2]
[1,]    1    0
[2,]    0    1
```

2.5 Quadratic forms

- A **quadratic form** for p variables $x_1, x_2, ..., x_p$ is a function of the form $\sum_{i=1}^{p}\sum_{j=1}^{p} a_{ij}x_ix_j$ which can be more compactly written as $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}$ with $\boldsymbol{A} = (a_{ij})$. Could you write $(x_1 + x_2)^2 = \sum_{ij}^{2} x_ix_j$ as $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}$?
- A symmetric matrix $\boldsymbol{A}$ is **positive definite** if $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x} > 0$ for every $\boldsymbol{x} \neq \boldsymbol{0}$. Is $\boldsymbol{A}$ above positive definite? Is $\boldsymbol{I}$?
- A symmetric matrix $\boldsymbol{A}$ is **positive semidefinite** if $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x} \geq 0$ for every $\boldsymbol{x} \neq \boldsymbol{0}$. Is $\boldsymbol{A}$ above positive semidefinite?

2.6 Determinant and trace

- The determinant of a square matrix $\boldsymbol{A}$ is defined by (for any i)

$$det(\boldsymbol{A}) = |\boldsymbol{A}| = \sum_{j=1}^{p}(-1)^{i+j}a_{ij}|\boldsymbol{A}_{-i,-j}|$$

  where $\boldsymbol{A}_{-i,-j}$ is the matrix found by removing the ith row and j'th column from $\boldsymbol{A}$ and |scalar| = scalar. Verify that $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$.

- The **determinant of a diagonal matrix** is $|diag(d_1, d_2, ..., d_p)| = \prod_{i=1}^{p} d_1 d_2 ... d_p$.
- A square matrix $\boldsymbol{A}$ is **positive definite** if and only if it is **nonsingular** if and only if $|\boldsymbol{A}| > 0$. If $|\boldsymbol{A}| = 0$, then $\boldsymbol{A}$ is singular. Compare $|\boldsymbol{J}_2|$ and $|\boldsymbol{I}_2|$.
- The **determinant of a product** of two conformable matrices is the product of their determinants:$|\boldsymbol{AB}| = |\boldsymbol{A}||\boldsymbol{B}|$. Compare $|\boldsymbol{J}_2\boldsymbol{I}_2|$ to $|\boldsymbol{J}_2||\boldsymbol{I}_2|$.

- The trace of a square matrix $\boldsymbol{A}$ is the sum of its diagonal elements: $tr(\boldsymbol{A}) = \sum_i a_{ii}$.
- $tr(\boldsymbol{A} + \boldsymbol{B}) = tr(\boldsymbol{A}) + tr(\boldsymbol{B})$. Consider $tr(\boldsymbol{J}_2 + 2\boldsymbol{I}_2)$.
- $tr(\boldsymbol{AB}) = tr(\boldsymbol{BA})$ when $\boldsymbol{AB}$ and $\boldsymbol{BA}$ are defined.

Code

```
> J=matrix(1,2,2)
> J
     [,1] [,2]
[1,]    1    1
[2,]    1    1
> I=diag(c(1,1))
> I
     [,1] [,2]
[1,]    1    0
[2,]    0    1
> det(I)
[1] 1
> det(J)
[1] 0
> det(I%*%J)
[1] 0
> sum(diag(I%*%J))
[1] 2
```

2.7 Orthogonality

- Vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ are **orthogonal** if $\boldsymbol{a}'\boldsymbol{b} = 0$.
- Orthogonal vectors are perpendicular. Plot $(0,1)'$ and $(1,0)'$. Are $(1,-1)'$ and $(1,1)'$ orthogonal?
- A vector $\boldsymbol{a}$ is **normalized** if $\boldsymbol{a}'\boldsymbol{a} = 1$. A vector is normalized by dividing by its length $\sqrt{\boldsymbol{a}'\boldsymbol{a}}$. How would you normalize $\boldsymbol{j}$?
- A matrix $\boldsymbol{C} = (\boldsymbol{c}_1, \boldsymbol{c}_2, ..., \boldsymbol{c}_p)$ is **orthogonal** if each of its columns are orthogonal to one another.
- If the columns of an orthogonal matrix $\boldsymbol{C}$ are also normalized, then $\boldsymbol{C}'\boldsymbol{C} = \boldsymbol{I}$. Interpretation: Transforming a vector or data point by multiplying by a normalized orthogonal matrix is referred to as a **rotation** about the origin. That is, if you take $\boldsymbol{z} = \boldsymbol{C}'\boldsymbol{x}$, then $\boldsymbol{z}$ and $\boldsymbol{x}$ are the same distance from the origin. Consider the point $(1,1)'$. Plot it and find its distance from the origin. Then multiply by the point by a normalized orthogonal matrix and find the new point's distance from the origin.

2.8 Eigenvalues and eigenvectors

- For a square matrix $\boldsymbol{A}$, a scalar λ, and nonzero vector $\boldsymbol{x}$ can be found satisfying
$$\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x},$$
or equivalently
$$(\boldsymbol{A} - \lambda\boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}.$$
- λ is called an **eigenvalue** and $\boldsymbol{x}$ is an **eigenvector**.
- Example: Verify that $\boldsymbol{J}_2$ has eigenvalue $\lambda_1 = 2$ and corresponding eigenvector $\boldsymbol{x}_1' = (1,1)$. Verify that it also has eigenvalue $\lambda_2 = 0$ and eigen vector $\boldsymbol{x}_2' = (1,-1)$
- In general, the **eigenvalues**, also called roots, can be found by finding the roots to the characteristic equation $|\boldsymbol{A} - \lambda\boldsymbol{I}|$.

- Find the eigenvalues of

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

- The **eigenvector** $\boldsymbol{x}$ corresponding to a particular eigenvalue λ_0 can be found by solving $(\boldsymbol{A} - \lambda_0 \boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}$.
- What are the eigenvectors for the eigenvalues above?
- Is each eigenvector unique?
- Unless otherwise stated, we will always use the normalized eigenvector.

Some useful properties:

- For a square matrix $\boldsymbol{A}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_p$,
  1. $tr(\boldsymbol{A}) = \sum_i \lambda_i$. What's $tr(\boldsymbol{A})$ from above?
  2. $|\boldsymbol{A}| = \prod_i \lambda_i$. What's $|\boldsymbol{A}|$ above?
  3. $rank(\boldsymbol{A})$ =number of non-zero eigenvalues. What's the rank of $\boldsymbol{A}$ above?
- Some results regarding positive definite and positive semi-definite matrices:
  1. All the eigenvalues of a positive definite matrix are positive.
  2. All nonzero eigenvalues of a positive semi-definite matrix are positive.

Spectral decomposition:

- If $\boldsymbol{A}$ is a symmetric and square matrix, there exists a normalized orthogonal matrix $\boldsymbol{C}$ such that

$$\boldsymbol{A} = \boldsymbol{C}\boldsymbol{D}\boldsymbol{C}'$$

  where $\boldsymbol{D} = diag(\lambda_1, \lambda_2, ..., \lambda_p)$ and the columns of $\boldsymbol{C}$ are the eigenvectors corresponding to $\lambda_1, \lambda_2, ..., \lambda_p$. The form in the above equation is called

the **spectral decomposition**. NOTE: By convention, we sort eigenvalues via $\lambda_1 \geq \lambda_2 ... \geq \lambda_p$. Also, make sure that the eigenvectors in $\boldsymbol{C}$ are in the same order as their corresponding eigenvalues.

- The **square root** of a positive definite symmetric matrix $\boldsymbol{A} = \boldsymbol{CDC}'$ can be defined $\boldsymbol{A}^{1/2} = \boldsymbol{CD}^{1/2}\boldsymbol{C}'$ where $\boldsymbol{D}^{1/2} = diag(\sqrt{\lambda_1}, \sqrt{\lambda_2}, ..., \sqrt{\lambda_p})$. Verify that $(\boldsymbol{A}^{1/2})^2 = \boldsymbol{A}$.

- We can compute $\boldsymbol{A}^{-1}$ or $\boldsymbol{A}^2$ in a similar way.

- Compare the spectral decomposition of

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

  to $\boldsymbol{J}_2$.

- The spectral decomposition of a covariance matrix is illuminating. Suppose that the above two matrices were covariance matrices, so that the off diagonal element was the covariance between two variables and the diagonal elements are the variances of each variable. How does the spectral decomposition describe the variability of the data?

Code

```
> A = matrix(c(1,.5,.5,1), 2, 2)
> A
     [,1] [,2]
[1,]  1.0  0.5
[2,]  0.5  1.0
> eigen(A)
$values
[1] 1.5 0.5

$vectors
          [,1]       [,2]
[1,] 0.7071068 -0.7071068
[2,] 0.7071068  0.7071068
```

```
> attach(eigen(A))
> values
[1] 1.5 0.5
> vectors
          [,1]       [,2]
[1,] 0.7071068 -0.7071068
[2,] 0.7071068  0.7071068

> vectors%*%diag(values)%*%t(vectors)
     [,1] [,2]
[1,]  1.0  0.5
[2,]  0.5  1.0
> det(A)
[1] 0.75
> prod(values)
[1] 0.75
> sum(diag(A))
[1] 2
> sum(values)
[1] 2
```

2.9 Singular value decompositions

The **singular value decomposition** of an $n \times p$ matrix $\boldsymbol{A}$ of rank $k \leq p$ is

$$\boldsymbol{A}_{n\times p} = \boldsymbol{U}_{n\times k}\boldsymbol{D}_{k\times k}\boldsymbol{V}'_{k\times p}$$

where

- $\boldsymbol{D} = diag(\lambda_1, \lambda_2, ..., \lambda_k)$ and λ_i^2 are the nonzero eigenvalues of $\boldsymbol{A}\boldsymbol{A}'$,
- The k columns of $\boldsymbol{U}$ are the normalized eigenvectors of $\boldsymbol{A}\boldsymbol{A}'$
- The k rows of of $\boldsymbol{V}'$ are the normalized eigenvectors of $\boldsymbol{A}'\boldsymbol{A}$.
- SVU is often used to plot $p > 2$ variables in two - dimensions (we take k=2), i.e. biplots are plots of $\boldsymbol{U}$ and $\boldsymbol{V}$.

Code

```
> Y
     [,1] [,2]
[1,]    1    4
[2,]    2    5
[3,]    3    6
> svd(Y)
$d
[1] 9.5080320 0.7728696

$u
           [,1]       [,2]
[1,] -0.4286671  0.8059639
[2,] -0.5663069  0.1123824
[3,] -0.7039467 -0.5811991

$v
           [,1]       [,2]
[1,] -0.3863177 -0.9223658
[2,] -0.9223658  0.3863177

> attach(svd(Y))
> u%*%diag(d)%*%v
     [,1] [,2]
[1,]    1    4
[2,]    2    5
[3,]    3    6
```

3 Describing multivariate data

Outline:

- Notations
- Mean vectors
- Covariance matrix
- Subsets of variables
- Linear combinations of variables
- Measures of overall variability

3.1 Notations

In univariate analysis, we denote a random variable with a capital letter Y and the i'th realization from this distribution by y_i. In multivariate analysis, we do not adopt this notation. Our notation is summarized below.

	univariate	multivariate
Random Variable	Y (1 dimensional)	**y** (p dimensional)
i'th Realization	y_i (scalar)	$\mathbf{y}_i$ ($p \times 1$ vector)
Entire Data Set	**y** ($n \times 1$ vector)	**Y** ($n \times p$ matrix)
j'th variable	—	y_j

- For **multivariate** data, p characteristics are measured for each experimental unit. Hence, we have data

$$\mathbf{y}_1 = \begin{pmatrix} y_{11} \\ y_{12} \\ \vdots \\ y_{1p} \end{pmatrix}, \mathbf{y}_2 = \begin{pmatrix} y_{21} \\ y_{22} \\ \vdots \\ y_{2p} \end{pmatrix}, \ldots, \mathbf{y}_n = \begin{pmatrix} y_{n1} \\ y_{n2} \\ \vdots \\ y_{np} \end{pmatrix}$$

- Our data matrix is

$$\mathbf{Y} = \begin{pmatrix} \mathbf{y}_1' \\ \mathbf{y}_2' \\ \vdots \\ \mathbf{y}_i' \\ \vdots \\ \mathbf{y}_n' \end{pmatrix} = \begin{pmatrix} y_{11} & y_{12} & \cdots & y_{1j} & \cdots & y_{1p} \\ y_{21} & y_{22} & \cdots & y_{2j} & \cdots & y_{2p} \\ \vdots & \vdots & & \vdots & & \vdots \\ y_{i1} & y_{i2} & \cdots & y_{ij} & \cdots & j_{ip} \\ \vdots & \vdots & & \vdots & & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nj} & \cdots & y_{np} \end{pmatrix}$$

Example 3.1. *Available soil calcium, exchangeable soil calcium and turnip green calcium were measured at 10 different turnip plots.*

Plot	y_1 available soil	y_2 exchangeable soil	y_3 turnip
1	35	3.5	2.8
2	35	4.9	2.70
3	40	30	4.8
4	10	2.8	3.21
5	6	2.7	2.73
6	20	2.8	2.81
7	35	4.6	2.88
8	35	10.9	2.9
9	35	8	3.28
10	30	1.6	3.20

- What are the experimental units?
- What are the variables?
- Describe y_{ij} for this data.
- What are $\mathbf{y}_1, \mathbf{y}_2, ...$?

Code

```
> turnip = read.table("T3_4_CALCIUM.dat", row.names=1,
 col.names=c("UNIT", "Available Soil", "Exchangeable Soil", "Turnip"))
> turnip[1:3,]
  Available.Soil Exchangeable.Soil Turnip
1             35               3.5   2.80
2             35               4.9   2.70
3             40              30.0   4.38
```

3.2 Mean vectors

- The **sample mean vector** of the data taken over the experimental units is

$$\bar{\mathbf{y}} = \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_p \end{pmatrix} = \frac{1}{n}\mathbf{Y}'\mathbf{j} = \sum_{i=1}^{n} \frac{1}{n}\mathbf{y}_i.$$

  or

$$\bar{\mathbf{y}}' = (\bar{y}_1, \bar{y}_2, ..., \bar{y}_p) = \frac{1}{n}\mathbf{j}'\mathbf{Y} = \sum_{i=1}^{n} \frac{1}{n}\mathbf{y}_i'$$

Code

```
> colMeans(turnip)
   Available.Soil Exchangeable.Soil            Turnip
           28.100             7.180             3.089
```

The **population mean** or expected value of the random vector of variables $\boldsymbol{y}$

is denoted

$$E(\boldsymbol{y}) = E\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{pmatrix} = \begin{pmatrix} E(y_1) \\ E(y_2) \\ \vdots \\ E(y_p) \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix} = \boldsymbol{\mu}.$$

- The sample mean vector will never be identical to the population mean vector. However, it is an **unbiased estimator** of $\boldsymbol{\mu}$ since $E(\bar{\mathbf{y}}) = \boldsymbol{\mu}$.

3.3 Covariance matrix

- The **population covariance** between two variables y_j and y_k, also called the pairwise covariance, is defined

$$\sigma_{jk} = Cov(y_j, y_k) = E[(y_j - \mu_j)(y_k - \mu_k)] = E[y_j y_k] - \mu_j \mu_k$$

- The **population covariance matrix** for a random vector $\boldsymbol{y} = (y_1, y_2, ..., y_p)'$ is a $p \times p$ matrix defined $Cov(\mathbf{y}) = \boldsymbol{\Sigma} = (\sigma_{jk})$. The elements are the covariances between y_j and y_k.

- We can compute $\boldsymbol{\Sigma}$ via $\boldsymbol{\Sigma} = E[(\mathbf{y} - \boldsymbol{\mu})(\mathbf{y} - \boldsymbol{\mu})'] = E[\mathbf{y}\mathbf{y}'] - \boldsymbol{\mu}\boldsymbol{\mu}'$. Note that the first expression for $\boldsymbol{\Sigma}$ can be used to verify that $\boldsymbol{\Sigma}$ is **positive definite or positive semi-definite**.

- The usual **sample covariance** between y_j and y_k is defined

$$s_{jk} = \frac{1}{n-1}\left(\sum_{i=1}^{n}(y_{ij} - \bar{y}_j)(y_{ik} - \bar{y}_k)\right) = \frac{1}{n-1}\left((\sum_{i=1}^{n} y_{ij}y_{ik}) - n\bar{y}_j\bar{y}_k\right)$$

- The **sample covariance matrix** is defined $\mathbf{S} = (s_{jk})$ and can be found by simply computing each s_{jk} or

$$\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})' = \frac{1}{n-1}\mathbf{Y}'(\mathbf{I} - \frac{1}{n}\mathbf{J})\mathbf{Y}$$

- The later expression is important for verifying that **S** is positive semi definite.
- The sample covariance is an **unbiased estimator** of σ_{jk} since $E(s_{jk}) = \sigma_{jk}$
- Use the sample covariance matrix for the turnip data to describe the relationship between variables. Does a scatter plot verify your interpretation?

Code

```
> cov(turnip)
                  Available.Soil Exchangeable.Soil    Turnip
Available.Soil        140.544444         49.680000 1.9412222
Exchangeable.Soil      49.680000         72.248444 3.6760889
Turnip                  1.941222          3.676089 0.2501211
> plot(turnip)
```

- The **population correlation matrix** is defined $\mathbf{P} = (\rho_{jk}) = \left(\frac{\sigma_{jk}}{\sqrt{\sigma_{jj}\sigma_{kk}}}\right)$
- ρ_{jj} = 1.
- ρ_{jk} near 1 means there is a strong positive linear relationship between y_j and y_k.
- ρ_{jk} near -1 means there is a strong negative linear relationship between y_j and y_k.
- ρ_{jk} near 0 means there is little linear relationship between y_j and y_k.
- Just as with covariance, two variables could be strongly related but have 0 correlation.

- The **sample correlation matrix** is defined $\mathbf{R} = (r_{jk}) = \left(\frac{s_{jk}}{\sqrt{s_{jj}s_{kk}}}\right)$

- For the turnip data, could you use **S** to compute **R**?
- Based on the scatter plots, do you think that correlation should be used to describes the strength and degree of the relationship between variables?

Code

```
> cov(turnip)
                  Available.Soil Exchangeable.Soil    Turnip
Available.Soil        140.544444         49.680000 1.9412222
Exchangeable.Soil      49.680000         72.248444 3.6760889
Turnip                  1.941222          3.676089 0.2501211
> cor(turnip)
                  Available.Soil Exchangeable.Soil   Turnip
Available.Soil         1.0000000         0.4930154 0.327411
Exchangeable.Soil      0.4930154         1.0000000 0.864762
Turnip                 0.3274110         0.8647620 1.000000
```

3.4 Subsets of variables

Sometimes it will be natural to group the variables into 2 groups of say p and q ($p + q$ total variables). We observe

$$\begin{pmatrix} \mathbf{y} \\ \mathbf{x} \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_p \\ x_1 \\ \vdots \\ x_q \end{pmatrix}$$

Example 3.2. *Suppose that you have 3 variables $\boldsymbol{z}' = (z_1, z_2, z_3)$, and you decide that z_1 and z_2 are naturally grouped. So you call $\boldsymbol{y} = (z_1, z_2)$ and $x_1 = z_3$. What would the partitioned data matrix of 4 experimental units look like?*

- The population mean vector is two mean vectors

$$E\begin{pmatrix} \mathbf{y} \\ \mathbf{x} \end{pmatrix} = \begin{pmatrix} E(\mathbf{y}) \\ E(\mathbf{x}) \end{pmatrix}$$

- The sample mean vector is two sample mean vectors

$$\begin{pmatrix} \bar{\mathbf{y}} \\ \bar{\mathbf{x}} \end{pmatrix} = \frac{1}{n}\sum_{i}^{n}\begin{pmatrix} \mathbf{y}_i \\ \mathbf{x}_i \end{pmatrix}$$

- The covariance matrix for two vectors $\mathbf{y}_p$ and $\mathbf{x}_q$ is

$$\mathbf{\Sigma}_{yx} = Cov(\mathbf{y}_p, \mathbf{x}_q) = (\sigma_{y_j x_k}) =$$

- The covariance matrix for a partitioned vector $\mathbf{z}' = (\mathbf{y}', \mathbf{x}')$ is

$$Cov\begin{pmatrix} \mathbf{y} \\ \mathbf{x} \end{pmatrix} = \begin{pmatrix} Cov(\mathbf{y},\mathbf{y}) & Cov(\mathbf{y},\mathbf{x}) \\ Cov(\mathbf{x},\mathbf{y}) & Cov(\mathbf{x},\mathbf{x}) \end{pmatrix} = \begin{pmatrix} \mathbf{\Sigma}_{yy} & \mathbf{\Sigma}_{yx} \\ \mathbf{\Sigma}_{xy} & \mathbf{\Sigma}_{xx} \end{pmatrix}$$

- The sample covariance matrix is

$$\mathbf{S} = \begin{pmatrix} \mathbf{S}_{yy} & \mathbf{S}_{yx} \\ \mathbf{S}_{xy} & \mathbf{S}_{xx} \end{pmatrix}$$

3.5 Example

Example 3.3. *Five measurements were taken on patients with diabetes. Three variables of major interest and two variables of minor interest are*
x_1 = *glucose intolerance*
x_2 = *insulin response to oral glucose*
x_3 = *insulin resistance*
y_1 = *relative weight*
y_2 = *fasting plasma glucose*

Data for 5 patients

Patient	y_1	y_2	x_1	x_2	x_3
1	0.81	80	356	124	55
2	0.95	97	289	117	76
3	0.94	105	319	143	105
4	1.04	90	356	199	108
5	1.00	90	323	240	143
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

- The sample covariance matrix and mean vector are

$$\begin{pmatrix} \bar{\mathbf{y}} \\ \bar{\mathbf{x}} \end{pmatrix} = \left(\begin{array}{c} 0.92 \\ 90.4 \\ \hline 340.8 \\ 171.4 \\ 97.8 \end{array}\right)$$

$$\begin{pmatrix} \mathbf{S}_{yy} & \mathbf{S}_{yx} \\ \mathbf{S}_{xy} & \mathbf{S}_{xx} \end{pmatrix} = \left(\begin{array}{cc|ccc} 0.02 & 0.22 & 0.79 & -0.21 & 2.19 \\ 0.22 & 70.56 & 26.23 & -23.96 & -20.84 \\ \hline 0.79 & 26.23 & 1106.41 & 396.73 & 108.38 \\ -0.21 & -23.96 & 396.73 & 2381.88 & 1142.64 \\ 2.19 & -20.84 & 108.38 & 1142.64 & 2136.40 \end{array}\right)$$

- Which sample covariance matrix describes the relationship between $\mathbf{y}$ and $\mathbf{x}$?
- Verify that $\mathbf{S}_{yx} = \mathbf{S}'_{xy}$. Will the same hold for $\mathbf{\Sigma}_{yx}$ and $\mathbf{\Sigma}_{xy}$? Will these properties be satisfied for sample and population correlation matrices?

Code

```
> diabetes= read.table("T3_5_DIABETES.dat",
+ col.names=c("Person","Y1","Y2","X1", "X2", "X3"), row.names=1)
>
> cov(diabetes)
            Y1          Y2           X1           X2           X3
```

```
Y1  0.01618184   0.216029    0.7871691   -0.2138454    2.189072
Y2  0.21602899  70.558937   26.2289855  -23.9560386  -20.841546
X1  0.78716908  26.228986 1106.4135266  396.7323671  108.383575
X2 -0.21384541 -23.956039  396.7323671 2381.8826087 1142.637681
X3  2.18907246 -20.841546  108.3835749 1142.6376812 2136.396135
> cor(diabetes)
            Y1          Y2         X1          X2          X3
Y1  1.00000000  0.20217252 0.18603532 -0.03444497  0.37231056
Y2  0.20217252  1.00000000 0.09387431 -0.05843578 -0.05368006
X1  0.18603532  0.09387431 1.00000000  0.24438735  0.07049590
X2 -0.03444497 -0.05843578 0.24438735  1.00000000  0.50653268
X3  0.37231056 -0.05368006 0.07049590  0.50653268  1.00000000
> y=diabetes[,1:2]
> x=diabetes[,3:5]
> cov(y,y)
           Y1        Y2
Y1 0.01618184  0.216029
Y2 0.21602899 70.558937
> cov(y,x)
           X1          X2         X3
Y1  0.7871691  -0.2138454   2.189072
Y2 26.2289855 -23.9560386 -20.841546
> cor(x,y)
            Y1          Y2
X1  0.18603532  0.09387431
X2 -0.03444497 -0.05843578
X3  0.37231056 -0.05368006
```

3.6 Linear combinations of variables

We will often be interested in linear combinations of our variables. For example, we reduce the dimensionality of the variables by taking some "optimal" linear combination of them. Contrasts are also an example of linear combinations of variables. In general, a linear transformation can be written

$$\mathbf{z} = \mathbf{A}\mathbf{y} + \mathbf{b}$$

3.6.1 Population properties

The following results hold for $\mathbf{a}$ a vector and b a scalar:

- $E[\mathbf{a}'\mathbf{y} + b] = \mathbf{a}'E[\mathbf{y}] + b$
- $Cov(\mathbf{a}'\mathbf{y} + b) = \mathbf{a}'Cov(\mathbf{z})\mathbf{a} = \mathbf{a}'\mathbf{\Sigma}\mathbf{a}$

Example 3.4. *Suppose* $\boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ *has some mean vector* $\boldsymbol{\mu}$ *and covariance matrix*

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$$

and consider transformation

$$z = \frac{1}{\sqrt{2}}\boldsymbol{j}'(\boldsymbol{y} - \boldsymbol{\mu}).$$

- What is $\mathbf{a}'$ above?
- What is $E[z]$?
- What is $Cov(z)$?
- How does $Cov(z)$ compare to $Cov(y_1)$ or $Cov(y_2)$? How does it compare to the first eigenvalue of $\mathbf{\Sigma}$?

The following results hold for $\mathbf{A}$ a matrix and $\mathbf{b}$ a vector

$$E(\mathbf{A}\mathbf{y} + \mathbf{b}) = \mathbf{A}E(\mathbf{y}) + \mathbf{b}$$

$$Cov(\mathbf{A}\mathbf{y} + \mathbf{b}) = \mathbf{A}\mathbf{\Sigma}\mathbf{A}'$$

Example 3.5. *Suppose* $\boldsymbol{y}$ *has some mean vector* $\boldsymbol{\mu}$ *and covariance matrix*

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$$

and consider transformation $\boldsymbol{z} = \boldsymbol{C}'(\boldsymbol{y} - \boldsymbol{\mu})$ *where* $\boldsymbol{C}$ *is a matrix of normalized eigenvectors for the spectral decomposition of* $\mathbf{\Sigma}$*, i.e.* $\mathbf{\Sigma} = \boldsymbol{CDC}'$

- What is $E[\mathbf{z}]$ and $Cov(\mathbf{z})$?

3.6.2 Sample properties

Suppose we have data matrix $\mathbf{Y} = (\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_n)'$ with mean vector $\bar{\mathbf{y}}$ and sample covariance $\mathbf{S}$. Consider the linear transformation for each $\mathbf{y}_i$

$$\mathbf{z}_i = \mathbf{A}\mathbf{y}_i + \mathbf{b}$$

and denote the matrix of transformed data $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_2, ..., \mathbf{z}_n)'$. Denote the mean vector $\bar{\mathbf{z}} = \frac{1}{n}\sum_i \mathbf{z}_i$ and covariance matrix $\mathbf{S}_z = \frac{1}{n-1}\sum_i (\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})'$. Then the following results hold:

$$\bar{\mathbf{z}} = \mathbf{A}\bar{\mathbf{y}} \text{ and } \mathbf{S}_z = \mathbf{A}\mathbf{S}\mathbf{A}'$$

Example 3.6. *Reconsider the turnip data.*

- What were $\bar{\mathbf{y}}$ and $\mathbf{S}$?

Code

```
> ybar = colMeans(turnip)
> S = cov(turnip)
> ybar
   Available.Soil Exchangeable.Soil            Turnip
           28.100             7.180             3.089
> S
                  Available.Soil Exchangeable.Soil    Turnip
Available.Soil        140.544444         49.680000 1.9412222
Exchangeable.Soil      49.680000         72.248444 3.6760889
Turnip                  1.941222          3.676089 0.2501211
```

- Note that the first normalized Eigen vector for $\mathbf{S}$ is $\mathbf{c}_1' = [0.884, 0.466, 0.021]$. Apply the transformation $z_i = \mathbf{c}_1'\mathbf{y}_i$ and use $\mathbf{Z}$ to get $\mathbf{S}_z$ and $\bar{z}$.

Code

```
> SpectralDecomposition=eigen(S)
> SpectralDecomposition
$values
[1] 166.75133359  46.23201359   0.05966281

$vectors
           [,1]        [,2]         [,3]
[1,] 0.88462536  0.46626967  0.005529478
[2,] 0.46584725 -0.88317582 -0.054651748
[3,] 0.02059895 -0.05092221  0.998490166

> c1 = SpectralDecomposition$vectors[,1]
> z = turnip[1]*c1[1] + turnip[2]*c1[2] + turnip[3]*c1[3]
> names(z) = "z"
> colMeans(z)
       z
28.26639
> var(z)
         z
z 166.7513
```

- Now compute $\mathbf{c}_1'\bar{\mathbf{y}}$ and $\mathbf{c}_1'\mathbf{S}\mathbf{c}_1$. How do these answers compare to above?

Code

```
> cov(z)
         z
z 166.7513
> t(c1)%*%S%*%c1
         [,1]
[1,] 166.7513
>
> colMeans(z)
       z
28.26639
> t(c1)%*%ybar
         [,1]
[1,] 28.26639
```

- Compare $\mathbf{S}_z$ to s_{11}, s_{22}, and s_{33}.

```
Code
> cbind(turnip, z)
   Available.Soil Exchangeable.Soil Turnip         z
1              35               3.5   2.80 32.650030
2              35               4.9   2.70 33.300156
3              40              30.0   4.38 49.450655
4              10               2.8   3.21 10.216749
5               6               2.7   2.73  6.621775
6              20               2.8   2.81 19.054762
7              35               4.6   2.88 33.164110
8              35              10.9   2.90 36.099359
9              35               8.0   3.28 34.756230
10             30               1.6   3.20 27.350033
> cov(cbind(turnip, z))
                  Available.Soil Exchangeable.Soil    Turnip          z
Available.Soil        140.544444         49.680000 1.9412222 147.512458
Exchangeable.Soil      49.680000         72.248444 3.6760889  77.680651
Turnip                  1.941222          3.676089 0.2501211   3.434903
z                     147.512458         77.680651 3.4349025 166.751334
```

- When units of measure are not comparable across variables, we often focus on centered and scaled data

$$\mathbf{z}_i = [diag(\mathbf{S})]^{-1/2}(\mathbf{y}_i - \bar{\mathbf{y}}).$$

- The covariance is equal to the correlation if the data are centered and scaled.

3.7 Measures of overall variability

We will be interested in measuring the overall variability of a data set $\mathbf{Y}$. The two main measures are $|\mathbf{S}|$ and $tr(\mathbf{S})$.

- How can these measures be expressed in terms of eigenvalues?
- One of these will be 0 when two or more variables are perfectly correlated. Which one?
- One of these does not take into account the correlation between variables. Which one?
- In general, large $|\mathbf{S}|$ or $tr(\mathbf{S})$ reflects large deviation of $\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_n$ from $\bar{\mathbf{y}}$. Sketch a scatter plot of correlated and uncorrelated data to better understand how the measures differ in describing the spread of the data. Recall for a 2x2 covariance matrix, $|\mathbf{S}|$ decreases as the off diagonal element gets farther from 0.

Code

```
> # load the MASS package
> library(MASS)
> A = mvrnorm(2000, c(0,0), matrix(c(1,.9,.9,1),2))
> B = mvrnorm(2000, c(0,0), matrix(c(1,0,0,1),2))
> plot(A,xlim = c(-3,3),ylim = c(-3,3),col = 3,ylab = "",xlab = "")
> par(new = T)
> plot(B,xlim = c(-3,3),ylim = c(-3,3),pch = 2,col = 2,ylab = "",xlab = "
> det(cov(A))
[1] 0.1855346
> det(cov(B))
[1] 1.058214
> sum(diag(cov(A)))
[1] 1.927741
> sum(diag(cov(B)))
[1] 2.058932
```

- What will be the determinant and trace of a $p \times p$ correlation matrix?
- Can you think of transformations that will in general preserve the overall variability? See the code below. What transformation is being applied?

Code

```
> S = cov(turnip)
> Sz = t(eigen(S)$vectors)%*%S%*%eigen(S)$vectors
> S = cov(turnip)
> A = eigen(S)$vectors
> A
           [,1]        [,2]         [,3]
[1,] 0.88462536  0.46626967  0.005529478
[2,] 0.46584725 -0.88317582 -0.054651748
[3,] 0.02059895 -0.05092221  0.998490166
> Sz = t(A)%*%S%*%A
> round(Sz, 2)
       [,1]  [,2] [,3]
[1,] 166.75  0.00 0.00
[2,]   0.00 46.23 0.00
[3,]   0.00  0.00 0.06
> det(S)
[1] 459.9555
> det(Sz)
[1] 459.9555
>
> sum(diag(S))
[1] 213.043
> sum(diag(Sz))
[1] 213.043
```

4 Multivariate normal distribution

Outline:

- Multivariate normal distribution
- Maximum likelihood estimation (MLE)
- Tests of normality

4.1 Multivariate normal distribution

The multivariate normal density of random vector $\mathbf{y}$ with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ is

$$f(\mathbf{y}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} e^{-\frac{1}{2}(\mathbf{y}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{y}-\boldsymbol{\mu})}$$

1. We often say that $\mathbf{y} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for short.
2. Compare this density to the univariate normal density.
3. We implicitly assume that $\boldsymbol{\Sigma}$ is full rank. What happens if it isn't?
4. The quantity $\Delta = (\mathbf{y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$ is called the Mahalanobis distance from $\mathbf{y}$ to $\boldsymbol{\mu}$. Will this distance always be positive? Why?
5. $|\boldsymbol{\Sigma}|$ is called the **generalized population variance**. It decreases as σ_{jk} increases ($j \neq k$) but increases if σ_{jj} increases.

Below are some important properties of MVN:

1. The family of MVN is closed under linear transformation. That is, for any $\mathbf{A}_{q\times p}$ matrix of rank q and $\mathbf{b}_q$ vector,

   $$\mathbf{A}\mathbf{y} + \mathbf{b} \sim N_q(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$$

   if $\mathbf{y}$ follows a MVN.

2. From above and our transformation results from the previous chapter,

$$\mathbf{z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \mathbf{I})$$

where $\boldsymbol{\Sigma}^{-1/2}$ a full rank square matrix such that $\boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}^{-1/2} = \boldsymbol{\Sigma}^{-1}$. How would you get $\boldsymbol{\Sigma}^{-1/2}$ using Eigen vectors and Eigen values?

3. We have $\Delta = (\mathbf{y} - \boldsymbol{\mu})'\Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu}) \sim \chi^2_p$.

4. Writing $\mathbf{y} = \begin{pmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{pmatrix}$, so that $\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}$ and $\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma_{11}} & \boldsymbol{\Sigma_{12}} \\ \boldsymbol{\Sigma_{21}} & \boldsymbol{\Sigma_{22}} \end{pmatrix}$. Then marginally, $\mathbf{y}_k \sim N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma_{kk}})$. This is actually a consequence of 1.

5. If $\boldsymbol{\Sigma}_{12} = \mathbf{0}$ above, then $\mathbf{y}_1$ and $\mathbf{y}_2$ are independent. Note that this result is not true for other distributions. This means that all dependent variables are linearly related.

6. If $\mathbf{y}$ and $\mathbf{x}$ are multivariate normal random vectors and independent, then provided their addition is defined,

$$\mathbf{y} + \mathbf{x} \sim N(\boldsymbol{\mu}_x + \boldsymbol{\mu}_y, \boldsymbol{\Sigma}_{xx} + \boldsymbol{\Sigma}_{yy})$$

7. If $\begin{pmatrix} \mathbf{y} \\ \mathbf{x} \end{pmatrix}$ has a MVN, then the conditional distribution of $\mathbf{y}|\mathbf{x}$ is a MVN with mean and covariance

$$E[\mathbf{y}|\mathbf{x}] = \boldsymbol{\mu_y} + \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{xx}^{-1}(\mathbf{x} - \boldsymbol{\mu_x}),\ Cov(\mathbf{y}|\mathbf{x}) = \boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}$$

To understand this better, consider the bivariate normal setting and regressing y on x. The usual model is $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ where $\epsilon_i \sim N(0, \sigma^2)$. This model implies that $y_i|x_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$.

   - Using the formulae, what is β_0 and β_1? Are the expressions familiar?
   - What is σ^2?

Summary of important results:

1. We can apply any linear transformation (rotations) to data without sacrificing normality.

2. Relationships between variables are linear for an MVN.

4.2 Maximum likelihood estimation (MLE)

Maximum likelihood estimators are the values of the parameters that maximize the likelihood. The **likelihood** of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ given a data set $\mathbf{Y}$ is defined

$$L(\boldsymbol{\mu}, \boldsymbol{\Sigma}; \boldsymbol{Y}) \;=\; \prod_{i=1}^{n} f(\boldsymbol{y}_i; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{np/2}|\boldsymbol{\Sigma}|^{n/2}} e^{-\frac{1}{2}\sum_{i=1}^{n}(\mathbf{y}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{y}_i-\boldsymbol{\mu})}$$

The maximum likelihood estimates of $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$ are as follows:

1. $\hat{\boldsymbol{\mu}} = \bar{\boldsymbol{y}}$.
2. $\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})' = \frac{n}{n-1}\mathbf{S}$
3. The MLE for correlation matrix $\mathbf{P}$ is $\hat{\mathbf{P}} = \mathbf{R}$.

Below are some properties of MLE:

- $\bar{\mathbf{y}} \sim N(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma})$ if $\mathbf{y}_i$ are i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. You can use property 6 to verify this.
- If $\mathbf{y}_i$ vary according to some multivariate distribution with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, then by the **Central Limit Theorem**,

  $$\sqrt{n}(\bar{\mathbf{y}} - \boldsymbol{\mu}) \xrightarrow{d} N(\mathbf{0}, \Sigma).$$

  Sometimes we say that $\bar{\mathbf{y}} \sim AN(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma})$ where "AN" is read asymptotically normal.
- $\mathbf{W} = \sum_{i=1}^{n}(\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})' = (n-1)\mathbf{S}$ varies according to a "Wishart" distribution. Its density is messy, but it is parameterized by its degrees of freedom (n-1), $\boldsymbol{\Sigma}$, and p. We write $\mathbf{W} \sim W_p(n-1, \boldsymbol{\Sigma})$. For univariate data, $W_1(n-1, 1)$ is the well-known χ^2_{n-1} distribution.
- $\bar{\mathbf{y}}$ and $\mathbf{W}$ are independent.

4.3 Test of normality

For univariate data, the most well-known method for assessing normality is via a **Q-Q plot**. A Q-Q plot is constructed by plotting the sample quantiles against the theoretical quantiles.

- If we order the data $y_{(1)} \leq y_{(2)} \ldots \leq y_{(n)}$, then $y_{(i)}$ corresponds to what sample quantile?
- How is the i/n'th theoretical quantile q_i defined? Draw a picture to explain.
- Why do you think we use the continuity corrected $(i-.5)/n$th theoretical quantiles instead of the i/nth quantile? What is the issue with the n/nth quantile?
- If q_i are the quantiles from a standard normal distribution and y_i are from a standard normal distribution, what will a plot of $(q_i, y_{(i)})$ look like?

Code

```
> z = rnorm(100)
> qqnorm(z)
> qqline(z)
```

- If y_i were drawn from a normal distribution, but not a standard normal distribution, then what would a plot of $(q_i, y_{(i)})$ look like?

Code

```
> x = 5 + 10*z
> qqnorm(x)
> qqline(x)
```

- What will a Q-Q plot look like if the data are heavy-tailed or skewed right?

Code
```
> x = rt(100, 2)
> qqnorm(x)
> qqline(x)
> x = rexp(100)
> qqnorm(x)
> qqline(x)
```

- Remark: Note that some software puts the theoretical quantiles on the Y axis, which changes the interpretation.

There are several approaches for assessing multivariate normality. We will illustrate these approaches on the ramus bone dataset. Recall it contains ramus bone measurements for 20 males at ages 8.5, 9, 9.5, and 10.

1. We can assess univariate normality since multivariate normality implies univariate normality. What's the weakness of this approach?

   - Q-Q plots for each variable.

Code
```
> bone = read.table('T3_7_BONE.dat', row.names=1, col.names=
+ c("Person","8.5", "9.0", "9.5", "10"))
> par(mfrow=c(2,2))
> qqnorm(bone[[1]], main="QQ plot: Age = 8.5")
> qqline(bone[[1]])
> qqnorm(bone[[2]], main = "QQ plot: Age = 9")
> qqline(bone[[2]])
> qqnorm(bone[[3]], main = "QQ plot: Age = 9.5")
> qqline(bone[[3]])
> qqnorm(bone[[4]], main = "QQ plot: Age = 10")
> qqline(bone[[4]])
```

- Is univariate normality satisfied? Why/why not?

- Q-Q plots on the PC scores.

```
> pcoutput<-princomp(bone)
> attach(pcoutput)
> sdev
   Comp.1    Comp.2    Comp.3    Comp.4
4.8787246 1.2869374 0.4556917 0.3287985
> par(mfrow = c(2,2))
> qqnorm(scores[,1], main = "QQ plot on PC 1")
> qqline(scores[,1])
> qqnorm(scores[,2], main = "QQ plot on PC 2")
> qqline(scores[,2])
> qqnorm(scores[,3], main = "QQ plot on PC 3")
> qqline(scores[,3])
> qqnorm(scores[,4], main = "QQ plot on PC 4")
> qqline(scores[,4])
```

  - Is normality supported?
  - Why might you be more concerned with the Q-Q plot of PC 2 than the plot of PC 4?
  - When will QQ plots on the PC scores work much better than Q-Q plots on the original data?

2. We can assess bivariate normality since multivariate normality implies bivariate normality. What's the weakness of this approach?

   - Create bi-variate scatter plots for each pair of variables.

```
> plot(bone)
```

  - Are the variables linearly related? Why would linear relationships be evidence in support of multivariate normality?

- Do you notice any outliers?

- Create a bi-variate scatter plot of the first two or three PC scores.

Code

```
> plot(scores)
> abline(v=0,h=0)
> summary(pcoutput)
Importance of components:
                          Comp.1     Comp.2      Comp.3      Comp.4
Standard deviation     4.8787246 1.28693742 0.455691682 0.328798463
Proportion of Variance 0.9234897 0.06425905 0.008056783 0.004194488
Cumulative Proportion  0.9234897 0.98774873 0.995805512 1.000000000
```

- Does the scatter plot support or refute multivariate normality?
- Is it reasonable to assess multivariate normality with the scatter plot? Why?
- If it were reasonable to assess multivariate normality with three PC scores but not two, what would you do?

3. Distance-based Q-Q plots. Recall that $\Delta = (\mathbf{y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu}) \sim \chi^2_p$. However, we can't compute these distances. Why?

   - We can compute the sample Mahalanobis distances $D_i = (\mathbf{y}_i - \bar{\mathbf{y}})'\mathbf{S}^{-1}(\mathbf{y}_i - \bar{\mathbf{y}})$ and compare these sample quantiles to quantiles from a χ^2_p distribution quantiles. Get a QQ plot of the Mahalonobis distances.

Code

```
### You have to install MVN first.  It's not a base package. ###
> library(MVN)
> Test=mvn(data = bone, mvn_test = "mardia")
> plot(Test,type='qq')
```

  - Is multivariate normality reasonable based on this plot?
  - Is the assumption that D_i has a χ^2_p distribution reasonable here?

- Gnanadesikan and Kettenring (1972) show that $u_i = \frac{nD_i}{(n-1)^2}$ has a beta distribution with shape parameters $a = 0.5p$ and $b = 0.5(n-p-1)$, which is well-known. Hence, if rigorous multivariate normality assessment is essential and n is small, we can construct a Q-Q plot of the u_is. Get the u_is and construct a Q-Q plot.

Code

```
> n=20
> p=4
> D = mahalanobis(bone,center=colMeans(bone), cov=cov(bone))
> U = n*D/(n-1)^2
> quantiles = qbeta((1:n - .5)/n, .5*p, .5*(n-p-1))
> qqplot(quantiles,U, main="QQ plot of U's", xlab="Sample Quantiles"
+ ylab="Theoretical Quantiles")
> abline(0,1)
```

  - Do you reach the same conclusion?

Mardia (1970) developed goodness-of-fit tests based on the skewness and kurtosis parameters for an MVN. The idea: For $\mathbf{x}$ and $\mathbf{y}$ two independent samples from the same p dimensional MVN the skewness parameter is

$$\beta_{1,p} = E[(\mathbf{y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})]^3 = 0$$

and the kurtosis parameter is

$$\beta_{2,p} = E[(\mathbf{y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})]^2 = p(p+2).$$

We can compare estimates of these parameters to their theoretical values under an MVN.

- Skewness test (skewed right or left)
  - Hypotheses: $H_0 : \beta_{1,p} = 0$ (not skewed) vs. $H_a : \beta_{1,p} \neq 0$ (skewed)

- Test statistic: $b_{1,p} = \frac{1}{n^2}\sum_i \sum_j D_{ij}^3$ where $D_{ij} = (\mathbf{y}_i - \bar{\mathbf{y}})'\mathbf{S}^{-1}(\mathbf{y}_j - \bar{\mathbf{y}})$
- Remark: For large samples (say $n > 50$) the test statistic is distributed according to a χ^2 distribution. For small sample sizes, a small sample correction is made in computing the p-value.

- Kurtosis test (thin or heavy tailed)
  - Hypotheses: $H_0 : \beta_{2,p} = p(p+2)$ (not thin or heavy tailed) vs. $H_a : \beta_{2,p} \neq p(p+2)$ (heavy or thin tailed)
  - Test statistic: $b_{2,p} = \frac{1}{n}\sum_i D_{ii}^2$.
  - Z score: $z = \frac{b_{2,p} - p(p+2)}{\sqrt{8p(p+2)/n}}$
  - Remark: Negative Z scores imply the kurtosis $b_{2,p}$ is too small. This means the data are thin-tailed. Thus, large Z-scores imply data are thick-tailed.
  - Remark: A small sample correction exists for this statistic and should be used if testing specifically for thin-tailed distributions by rejecting H_0 for small Z-scores. It uses $z = \frac{b_{2,p} - p(p+2)\left(\frac{n+p+1}{n}\right)}{\sqrt{8p(p+2)/(n-1)}}$

- Application: Get the p-values and Z-scores for each test above for the bone data.

Code

```
> library(MVN)
> mvn(data = bone, mvn_test = "mardia")
> p=4
> n=20
> b2p = 28.88
> (b2p - p*(p+2)*(n+p+1)/n)/sqrt(8*p*(p+2)/(n-1))
[1] -0.3523256
```

  - Are the data skewed? Why/why not?
  - Are the data thin or heavy-tailed? Why/why not?

Box and Cox(1964) proposed the class of transformations for data y:

$$y^{(\lambda)} = \begin{cases} \frac{y^{\lambda}-1}{\lambda} & \lambda \neq 0 \\ log(y) & \lambda = 0 \end{cases} \tag{1}$$

- The important quantity here is λ. Note that subtracting 1 and dividing by λ does not affect the normality, but is done so that the transformation is a smooth function of λ (this allows for us to find the MLE).
- Common values of λ are $-2, -1, 0, 1, 2$. What is the transformation for these?
- We can get maximum likelihood estimates of λ via a grid search:
  1. For each $\lambda = -2, -1.9, ..., 1.9, 2$, make the transformation, get MLEs of μ_λ and σ^2_λ on the transformed data and record the likelihood $l(\lambda)$.
  2. The MLE of λ is the value that maximizes $l(\lambda)$.

Andrews, Gnadadesikan, and Warner (1971) consider the extension to the multivariate normal approach. We now transform each variable as in the univariate approach via $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, ..., \lambda_p)'$

- MLE of $\boldsymbol{\lambda}$ could be found by performing a grid search over $\boldsymbol{\lambda}$.
- Sometimes, more complex but more efficient numerical approaches to MLEs fail when finding maxima of more than one parameter simultaneously, especially when data are highly correlated. In practice, finding the univariate MLE of λ_i one at a time yields a transformation similar to above.
- Confidence intervals for λ_i give us a range of reasonable transformations. These are based on the fact that MLEs are asymptotically normal. What should we do if 1 is contained in the confidence interval for λ_i?
- We can also see if the data needs to be transformed by testing the null hypothesis $H_0 : \lambda_i = 1$ against the alternative $H_a : \lambda_i \neq 1$. Note that the p-value for testing $H_0 : \lambda_i = 1$ is greater than 0.05 if 1 is contained in the confidence interval for λ_i.

- Let's see if the bone data needs to be transformed.
  - What happens to the powerTransform function when applied to the bone data?

Code

```
> library(car)
> summary(powerTransform(bone))
bcPower Transformations to Multinormality

     Est.Power Std.Err. Wald Lower Bound Wald Upper Bound
X8.5   -0.6533   3.2383          -7.0005           5.6938
X9.0   -1.2935   2.4839          -6.1619           3.5750
X9.5   -3.9624   1.4962          -6.8949          -1.0298
X10    -6.2307   0.3912          -6.9975          -5.4638

Likelihood ratio tests for transformation parameters
                                       LRT df      pval
LR test, lambda = (0 0 0 0)      2.630818  4 0.6213736
LR test, lambda = (1 1 1 1)      3.678809  4 0.4512155
LR test, lambda = (1 1 -2 -6.23) 1.359984  4 0.8511193
Warning message:
In estimateTransform.default(X, Y, weights, family, start, method,
  Convergence failure: return code = 52
```

  - Run powerTransform on the first couple of variables.

Code

```
 > summary(powerTransform(bone[[1]]))
bcPower Transformation to Normality

          Est.Power Std.Err. Wald Lower Bound Wald Upper Bound
bone[[1]]   -2.9333   4.9099         -12.5567           6.6901

Likelihood ratio tests about transformation parameters
                             LRT df      pval
LR test, lambda = (0) 0.3579905  1 0.5496244
LR test, lambda = (1) 0.6439369  1 0.4222888
> summary(powerTransform(bone[[2]]))
```

```
bcPower Transformation to Normality

          Est.Power Std.Err. Wald Lower Bound Wald Upper Bound
bone[[2]]   -2.5172   4.6323         -11.5965           6.5622

Likelihood ratio tests for transformation parameters
                            LRT df      pval
LR test, lambda = (0) 0.2943014  1 0.5874772
LR test, lambda = (1) 0.5736913  1 0.4487958
```

* Should you transform the data?

In univariate analysis, outliers can affect the mean and/or variance estimates. Recall that in multivariate analysis, an outlier can affect the mean, variance, and/or correlation estimates.

- Sketch a bivariate scatter plot where the outlier will affect the mean and variance but not the correlation.
- Sketch a bivariate scatter plot where the outlier will affect only the correlation.

- Construct boxplots of the original data or the first couple of PC scores.
- Construct scatter plots of the original data or the first couple of PC scores.
- Application: Get boxplots and scatter plots of the first couple of PC scores.

Code

```
> boxplot(bone)
> boxplot(scores)
> plot(scores)
> abline(h = 0, v = 0)
```

- Are the outliers affecting the mean, variance, or correlation?
- What would you expect boxplots of the original data and PC scores to look like if the outlier was affecting the mean and variance but not the correlation?

We can more formally detect outliers by comparing each D_i to the upper $0.05/n$th or $0.10/n$th quantiles from a chi-squared distribution. We can also compare the u_is to these quantiles from a beta distribution.

- Application: For the bone data with $n = 20$ and $p = 4$, the upper $0.05/20$ and $0.10/20$ quantiles from a beta distribution with shape parameters $0.5p = 2$ and $0.5(n - p - 1) = 7.5$ are 0.64 and 0.60. The upper 0.05/20 and 0.10/20 quantiles from a χ^2_4 distribution are 16.4 and 14.4.

Code

```
> qchisq(.05/n,p, lower.tail=F)
[1] 16.42394
> qbeta(.05/n,.5*p,.5*(n-p-1), lower.tail=F)
[1] 0.6444192
>
> round(cbind(scale(bone)[,1:4], D,U),2)
    X8.5  X9.0  X9.5   X10     D    U
1  -0.34 -0.32 -0.60 -0.64  0.76 0.04
2  -0.90 -0.92 -1.09 -1.12  1.30 0.07
3  -0.94 -1.11 -1.05 -1.08  1.76 0.10
4  -1.41 -1.70 -1.70 -1.56  3.85 0.21
5  -0.42 -0.44 -0.63 -0.79  0.87 0.05
6   1.53  1.41  1.04  0.83  2.81 0.16
7   1.01  1.33  1.42  1.08  4.29 0.24
8   0.46  0.15 -0.10  0.46  7.99 0.44
9  -0.22  0.46  0.66  1.08 11.03 0.61
10 -1.45 -1.03 -1.24 -1.15  5.85 0.32
11  1.01  0.70  0.39  0.17  2.83 0.16
12 -0.06 -0.17  0.92  1.49 10.57 0.59
13  1.37  1.25  1.19  1.30  2.59 0.14
14 -0.18 -0.29 -0.48 -0.60  0.66 0.04
15  0.38  0.31  0.24  0.13  0.32 0.02
16  0.81  0.82  0.81  0.68  0.83 0.05
17 -0.58 -0.76 -0.83 -0.71  1.11 0.06
```

```
18  1.85  1.96  1.72  1.41  4.36 0.24
19 -0.98 -0.84 -0.94 -1.12  2.11 0.12
20 -0.94 -0.80  0.28  0.13 10.09 0.56
```

- Do you detect any outliers? Why/why not? Are there any borderline outliers?
- Describe the nature of these borderline outliers using the centered and scaled data.
- How does your normality assessment affect boxplot or distance-based outlier analysis?

5 Inference on multivariate normal distribution

Outline

- Tests on one mean vector
- Comparing two mean vectors
- Follow-up testing
- Profile analysis
- Permutation tests

5.1 Tests on one mean vector

Example 5.1. *Recall the turnip data, in which three variables were measured on 10 different soil samples from a turnip farm. The variables were*

y_1 = *available soil calcium*
y_2 = *exchangeable soil calcium*
y_3 = *turnip green calcium*

Ag science research tells us that the desired calcium levels for quality turnip growth are (15, 6, 2.85). Recall that our sample mean vector and covariance matrix were

$$\bar{\boldsymbol{y}} = \begin{pmatrix} 28.1 \\ 7.18 \\ 3.09 \end{pmatrix} \text{ and } \boldsymbol{S} = \begin{pmatrix} 140.5 & 49.7 & 1.9 \\ 49.7 & 72.3 & 3.7 \\ 1.9 & 3.7 & .25 \end{pmatrix}$$

- Is $\bar{\mathbf{y}}$ too far from the optimal value $\boldsymbol{\mu}_0 = (15, 6, 2.85)'$ to believe that the farm is optimal? How would you measure the distance?

- What is $E[\bar{\mathbf{y}}]$ and $Cov(\bar{\mathbf{y}})$ if the turnip farm is optimal?

- Assuming for the moment that $\mathbf{\Sigma}$ is known. What is the distribution of Δ under the null hypothesis

$$\Delta = (\bar{\mathbf{y}} - \boldsymbol{\mu}_0)' \left(\frac{\mathbf{\Sigma}}{n}\right)^{-1} (\bar{\mathbf{y}} - \boldsymbol{\mu}_0)$$

- What are reasonable values of Δ under H_0? What are unreasonable values of Δ under H_0?
- Is $\mathbf{\Sigma}$ known?

Hypothesis testing:

- Model: $\boldsymbol{y}_i \overset{i.i.d.}{\sim} N(\boldsymbol{\mu}, \mathbf{\Sigma})$
- Hypotheses: $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ vs $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$
- Hotellings T^2 test statistic:

$$T^2 = n(\bar{\mathbf{y}} - \boldsymbol{\mu}_0)'\mathbf{S}^{-1}(\bar{\mathbf{y}} - \boldsymbol{\mu}_0) = (\bar{\mathbf{y}} - \boldsymbol{\mu}_0)' \left(\frac{\mathbf{S}}{n}\right)^{-1} (\bar{\mathbf{y}} - \boldsymbol{\mu}_0)$$

- Distribution of T^2:
  - Under the MVN model and under $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$, T^2 has a Hotellings T^2 distribution depending on parameter p and degrees of freedom $v = n - 1$. For short, we write $T^2_{p,v}$ to refer to this random variable and write $T^2_{\alpha,p,v}$ to refer to upper α quantiles. See Table A.7.
  - The $T^2_{p,v}$ distribution is closely related to the F distribution.

$$\frac{v - p + 1}{vp} T^2_{p,v} = F_{p,v-p+1}$$

- Level α decision rules: A **level** α test rejects H_0 if
  1. $T^2 > T_{\alpha,p,v}$ or
  2. $F > F_{\alpha,p,v-p+1}$ or

3. p-value $< \alpha$ where the p-value is $\Pr(F_{p,v-p+1} > F)$

Critical thinking:

- How is this test statistic similar to Δ?
- Why are large values of T^2 evidence against H_0?
- Why is a "one sided" test not well defined?
- The model assumes data have an MVN distribution. Do you think this assumption is important for large n?
- What happens if you try to perform the test and $v < p$?

Code

```
> turnip = read.table('T3_4_CALCIUM.dat', row.names=1,
 + col.names=c("UNIT", "Available Soil", "Exchangeable Soil", "Turnip"))
> ###### First get the mean, covariance, n, p, v and mu0 #####
> S=cov(turnip)
> ybar=colMeans(turnip)
> n=10
> v=n-1
> p=3
> mu0=c(15,6,2.85)
> #### Below are two ways to get T^2 ####
> Tsq = n*t((ybar - mu0))%*%solve(S)%*%(ybar-mu0)
> Tsq
         [,1]
[1,] 24.55891
> Tsq = mahalanobis(ybar, mu0, S/n)
> Tsq
[1] 24.55891
> F = (v - p +1)/(v*p) * Tsq
> F
[1] 6.367124
>
> qf(.99, p, v-p+1)
[1] 8.451285
```

```
>
> p.value=1-pf(F, p, v-p + 1)
> p.value
[1] 0.02068015
> plot(turnip)
> biplot(princomp(turnip, cor = T))
```

- Can you reject H_0 at the $\alpha = 0.01$ level? What about at the $\alpha = 0.05$ level?
- Assess the test.

5.2 Comparing Two Mean Vectors

Example 5.2. *Four different psychological tests were given to male and female "professional" job applicants. For each test and each person, the number of correct answers is recorded. The data are presented in Table 1. The tests are*

y_1 = *pictorial inconsistencies*

y_2 = *paper form board*

y_3 = *tool recognition*

y_4 = *vocabulary.*

- The goal is to determine if male job applicants are psychologically different from female job applicants.

Males					Females				
Person	Y1	Y2	Y3	Y4	Person	Y1	Y2	Y3	Y4
1	15	17	24	14	1	13	14	12	21
2	17	15	32	26	2	14	12	14	26
3	15	14	29	23	3	12	19	21	21
4	13	12	10	16	4	12	13	10	16
5	20	17	26	28	5	11	20	16	16
6	15	21	26	21	6	12	9	14	18
7	15	13	26	22	7	10	13	18	24
8	13	5	22	22	8	10	8	13	23
9	14	7	30	17	9	12	20	19	23
10	17	15	30	27	10	11	10	11	27
11	17	17	26	20	11	12	18	25	25
12	17	20	28	24	12	14	18	13	26
13	15	15	29	24	13	14	10	25	28
14	18	19	32	28	14	13	16	8	14
15	18	18	31	27	15	14	8	13	25
16	15	14	26	21	16	13	16	23	28
17	18	17	33	26	17	16	21	26	26
18	10	14	19	17	18	14	17	14	14
19	18	21	30	29	19	16	16	15	23
20	18	21	34	26	20	13	16	23	24
21	13	17	30	24	21	2	6	16	21
22	16	16	16	16	22	14	16	22	26
23	11	15	25	23	23	17	17	22	28
24	16	13	26	16	24	16	13	16	14
25	16	13	23	21	25	15	14	20	26
26	18	18	34	24	26	12	10	12	9
27	16	15	28	27	27	14	17	24	23
28	15	16	29	24	28	13	15	18	20
29	18	19	32	23	29	11	16	18	28
30	18	16	33	23	30	7	7	19	18
31	17	20	21	21	31	12	15	7	28
32	19	19	30	28	32	6	5	6	13

Table 1: Test scores of men and women on 4 different tests.

Hypothesis testing:

- Model: Assume that
  - $\mathbf{y}_{11}, \mathbf{y}_{12}, ..., \mathbf{y}_{1n_1} \overset{i.i.d.}{\sim} N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathbf{y}_{21}, \mathbf{y}_{22}, ..., \mathbf{y}_{2n_2} \overset{i.i.d.}{\sim} N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$
  - $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2$
  - data are independently sampled across populations
- Hypotheses: $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ vs. $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$
- Test statistic:
  - the sample mean and covariance vectors for populations $i = 1, 2$:

$$\bar{\mathbf{y}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{y}_{ij} \text{ and } \mathbf{S}_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\mathbf{y}_{ij} - \bar{\mathbf{y}}_i)(\mathbf{y}_{ij} - \bar{\mathbf{y}}_i)'$$

  - the pooled sample covariance:

$$\mathbf{S}_p = \frac{(n_1 - 1)\mathbf{S}_1 + (n_2 - 1)\mathbf{S}_2}{n_1 + n_2 - 2}$$

  - Hotellings test statistic:

$$T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{y}}_1 - \bar{\mathbf{y}}_2)' \mathbf{S}_p^{-1} (\bar{\mathbf{y}}_1 - \bar{\mathbf{y}}_2) = (\bar{\mathbf{y}}_1 - \bar{\mathbf{y}}_2)' \left[\mathbf{S}_p \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \right]^{-1} (\bar{\mathbf{y}}_1 - \bar{\mathbf{y}}_2)$$

- Distribution of T^2:
  - Under the model and H_0, T^2 has a Hotelling's T^2 distribution with parameters p and $v = n_1 + n_2 - 2$. See table A.7.
  - We can still get and F statistic using $F = \frac{v-p+1}{vp} T^2$ and p-value via $\Pr(F_{p,v-p+1} > F)$
- Level α decision rules: A **level** α test rejects H_0 if
  1. $T^2 > T_{\alpha,p,v}$ or
  2. $F > F_{\alpha,p,v-p+1}$ or

3. p-value $< \alpha$ where the p-value is $\Pr(F_{p,v-p+1} > F)$

Critical thinking:

- Could this test be motivated as a Mahalanobis distance?
- Why are large values of T^2 evidence against H_0?
- Why is a "one sided" test not well defined?
- The model assumes data have an MVN distribution. Do you think this assumption is important for large n_1 and n_2?
- The model also assumes that the covariances from each population are equal. When do you think this assumption is important?
- What happens if you try to perform the test and $v < p$?
- Both of Hotellings' T^2 tests are equivalent to a likelihood ratio test, and likelihood ratio tests are known to be asymptotically optimal. What do you think a likelihood ratio test is?
- There is an additional assumption that data are sampled independently (within and across populations for the two sample test). This assumption is typically only verified with statistical tools if data are collected sequentially. It is more often verified by a careful description of the manner in which data are generated. Can you think of studies where this assumption might not be satisfied?
- Why not add a bunch of redundant variables? After all, T^2 must increase if you add an additional variable. So won't the power also increase?

Let's revisit our psychological data.

- Write out the statistical model and state the hypotheses.

Code

```
> psycho=read.table("T5_1_PSYCH.dat",
+ col.names = c("Gender", "Y1","Y2","Y3","Y4"))
>
> males=psycho[1:32,-1]
> females = psycho[33:64,-1]
>
> biplot(princomp(males))
> biplot(princomp(females))
> n1=32
> n2=32
> S1=cov(males)
> S2=cov(females)
> ybar1=colMeans(males)
> ybar2=colMeans(females)
>
> S1
         Y1        Y2        Y3        Y4
Y1 5.192540  4.545363  6.522177  5.250000
Y2 4.545363 13.184476  6.760081  6.266129
Y3 6.522177  6.760081 28.673387 14.467742
Y4 5.250000  6.266129 14.467742 16.645161
> S2
         Y1        Y2        Y3        Y4
Y1 9.136089  7.549395  4.863911  4.151210
Y2 7.549395 18.603831 10.224798  5.445565
Y3 4.863911 10.224798 30.039315 13.493952
Y4 4.151210  5.445565 13.493952 27.995968
>
> sum(diag(S1))
[1] 63.69556
> sum(diag(S2))
[1] 85.7752
```

- Are the assumptions reasonable?

Code

```
> Sp = ((n1 - 1)*S1 + (n2-1)*S2)/(n1+n2 - 2)
> T.sq = n1*n2/(n1+n2)*t(ybar1 - ybar2)%*%solve(Sp)%*%(ybar1-ybar2)
> T.sq = mahalanobis(ybar1, ybar2, cov = Sp*(1/n1 + 1/n2))
> ### Transform to F stat ###
> p=4
> v=n1+n2-2
> F = (v-p+1)/(v*p)*T.sq
> F
[1] 23.21971
> qf(.95,p,v-p+1)
[1] 2.527907
>
> 1-pf(F,p,v-p+1)
[1] 1.464318e-11
> ### Use the Hotelling package to get the p-value ###
> library(Hotelling)
>
> print(hotelling.test(males,females))
Test stat:  97.601
Numerator df:  4
Denominator df:  59
P-value:  1.464e-11
>
> print(hotelling.test(Y1 + Y2 + Y3 + Y4~Gender, data = psycho))
Test stat:  97.601
Numerator df:  4
Denominator df:  59
P-value:  1.464e-11
```

- Perform the test and interpret the results.

Matched pairs:

Example 5.3. *To compare two types of coating for pipes, for each of 15 pairs of pipes, a pipe was given one of two types of coatings. Each pair of pipes was buried at a different location for 5 months. They were then dug up, and the corrosion of each pipe was measured with*

y_1 = *maximum depth of a "pit" (in thousandths of an inch), and*

y_2 = *number of pits.*

	Coating 1		Coating 2	
Pair	y_1	y_2	x_1	x_2
1	73	331	51	35
2	43	19	41	14
3	47	30	32	26
⋮	⋮	⋮	⋮	⋮
15	75	18	68	13

- What are the experimental units?
- We would like to see if coating types are different. How would you analyze this type of data if you had only taken one measurement per pair?
- How would you test the null hypothesis that the coatings were the same using the new variables $d_1 = y_1 - x_1$ and $d_2 = y_2 - x_2$?

5.3 Follow-up testing

Now that we have rejected $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ what can we say about each of the univariate null hypotheses $H_{0r} : \mu_{1r} = \mu_{2r}$ for $r = 1, 2, ..., p$?
Critical thinking:

1. Does the rejection of the multivariate null hypothesis mean that $\mu_{1r} \neq \mu_{2r}$ for every r?

2. Though we have declared that $\mu_{1r} \neq \mu_{2r}$ for some r, it need not be the case that we reject any of the univariate hypothesis tests. Why?

3. The **experiment-wise error rate** or the family-wise error rate (FWER) is the probability that 1 or more null hypotheses are rejected in error during the analysis of the experiment. If we construct confidence intervals, we should ensure that the probability that a confidence interval "captures" the true value of $\mu_{1r} - \mu_{2r}$ **for all of the** p **intervals** is $1 - \alpha$. Why do we control the experimentwise error rate or construct 100(1-α) simultaneous confidence intervals?

4. Can you think of an approach that would ensure that the experiment-wise error rate is controlled if performing a hypothesis test on each of the p variables? Is this approach conservative?

5. Hummel and Sligo (1971) argued that if the **the multivariate null hypothesis is rejected** at level α, then individual univariate inference for each variable can be made at level α. They argue that this **protected** approach controls the experimentwise error rate at level α. See Table 5.2 for an assessment of this approach.

   - Do you think this approach is reasonable if p is large, say 100 or 1000?
   - Compare this approach to a Bonferroni approach in terms of FWER control and power or confidence interval width.

Recall for the psychological data that we reject our multivariate null hypothesis at level 0.05. Report and interpret the appropriate confidence intervals for comparing each individual variable across populations in follow-up analysis.

Code

```
> t.test(males[1],females[1], var.equal=T)

        Two Sample t-test

data:  males[1] and females[1]
t = 5.4173, df = 62, p-value = 1.047e-06
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 2.287376 4.962624
sample estimates:
mean of x mean of y
 15.96875  12.34375

> t.test(males[1],females[1], var.equal=T, conf.level=1-.05/4)

        Two Sample t-test

data:  males[1] and females[1]
t = 5.4173, df = 62, p-value = 1.047e-06
alternative hypothesis: true difference in means is not equal to 0
98.75 percent confidence interval:
 1.903487 5.346513
sample estimates:
mean of x mean of y
 15.96875  12.34375

>
> t.test(males[2],females[2], var.equal=T)

        Two Sample t-test

data:  males[2] and females[2]
t = 2.0066, df = 62, p-value = 0.04915
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.007651476 3.992348524
```

```
sample estimates:
mean of x mean of y
 15.90625  13.90625

> t.test(males[2],females[2], var.equal=T, conf.level=1-.05/4)

        Two Sample t-test

data:  males[2] and females[2]
t = 2.0066, df = 62, p-value = 0.04915
alternative hypothesis: true difference in means is not equal to 0
98.75 percent confidence interval:
 -0.56414  4.56414
sample estimates:
mean of x mean of y
 15.90625  13.90625
```

5.4 Profile analysis

One sample profile analysis

Example 5.4. *For the psychological data, recall that for 32 women and 32 men, we counted the number of correct answers on an exam concerning*

y_1 = *pictorial inconsistencies*
y_2 = *paper form board*
y_3 = *tool recognition*
y_4 = *vocabulary*

Suppose that we wanted to test the null hypothesis that men scored the same on every test.

- How would you write this hypothesis in terms of $\mu_{11}, \mu_{12}, \mu_{13}$, and μ_{14}?
- Sketch a plot of (r, μ_{1r}) under H_0. This is called a **profile plot**.

- Could you write the null hypothesis via $H_0 : \mathbf{C}\boldsymbol{\mu}_1 = \mathbf{0}$ for some matrix $\mathbf{C}$?
- Now note that we can transform our variables via $\mathbf{z} = \mathbf{C}\mathbf{y}_1$ and test $H_0 : \boldsymbol{\mu}_z = \mathbf{0}$ against $H_1 : \boldsymbol{\mu}_z \neq \mathbf{0}$. We still use T^2 to test the null hypothesis, we are just applying the T^2 to the transformed data. That is we have

$$T^2 = n(\bar{\mathbf{z}} - \mathbf{0})'\mathbf{S}_z^{-1}(\bar{\mathbf{z}} - \mathbf{0}).$$

  Denoting the covariance matrix and mean vector of males by $\mathbf{S}_1$ and $\bar{\mathbf{y}}_1$, what is the test statistic in terms of $\mathbf{S}_1$ and $\bar{\mathbf{y}}_1$ and $\mathbf{C}$?
- What are v and p?
- Does the MVN assumption need to be assessed again on the transformed data?
- Formally state and test the null hypothesis

Code

```
> C = matrix(c(1, -1, 0, 0, 1, 0, -1, 0, 1, 0, 0, -1),nrow=3, byrow = T)
> ## Two ways to get T^2 ####
> n1*t(C%*%ybar1)%*%solve(C%*%S1%*%t(C))%*%C%*%ybar1
         [,1]
[1,] 221.1263
> z = as.matrix(males)%*%t(C)
> mahalanobis(colMeans(z),c(0,0,0), cov = cov(z)/n1)
[1] 221.1263
```

Critical thinking:

- For one sample profile analysis, we use $p = rank(\mathbf{C})$ and $v = n-1$. Why?
- Why must $\mathbf{C}$ be of full rank?

- Is **C** unique in the problem above?

Code
```
> C = matrix(c(1,-1,0,0,  0,1,-1,0, 0,0,1,-1),nrow=3, byrow = T)
> ## Two ways to get T^2 ####
> n1*t(C%*%ybar1)%*%solve(C%*%S1%*%t(C))%*%C%*%ybar1
         [,1]
[1,] 221.1263
```

Orthogonal polynomial contrasts

- For the bone data, where recall ramus bone measurements were taken on males aged 8.5, 9, 9.5, and 10, you could consider the following **C** matrix. What does each row represent? Plot the coefficients for each row.

```
-3 -1  1 3
 1 -1 -1 1
-1  3 -3 1
```

- Why are the above called orthogonal polynomial contrasts?

Code
```
> C = matrix(c(-3,-1,1,3, 1,-1,-1,1, -1,3,-3,1), byrow = T, nrow=3)
> z = as.matrix(bone)%*%t(C)
> p=3
> n=20
> v=n-1
> T.sq = mahalanobis(colMeans(z),c(0,0,0), cov = cov(z)/n)
> F = (v-p+1)/(v*p)*T.sq
> 1-pf(F,p,v-p+1)
[1] 2.616668e-06
```

- Is there evidence of a trend for the bone data? State the null hypothesis formally, and in words, interpret the results.

Code

```
> t.test(z[,1])

        One Sample t-test

data:  z[, 1]
t = 7.1547, df = 19, p-value = 8.445e-07
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
  6.589993 12.040007
sample estimates:
mean of x
    9.315

> t.test(z[,2])

        One Sample t-test

data:  z[, 2]
t = -0.44537, df = 19, p-value = 0.6611
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.5414508  0.3514508
sample estimates:
mean of x
   -0.095

> t.test(z[,3])

        One Sample t-test

data:  z[, 3]
t = -0.10549, df = 19, p-value = 0.9171
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.9378468  0.8478468
sample estimates:
mean of x
   -0.045
```

- What type of trend is detected?

- See table A.13 for **C** matrices for orthogonal polynomial contrasts when $p = 2, 3, ...10$.

Two sample profile analysis

Example 5.5. *Suppose that you wish to determine if the difference between men and women depends on test type.*

- Sketch several profile plots of (r, μ_{1r}) and (r, μ_{2r}) that would correspond to the null hypothesis that the difference between men and women does not depend on the test.
- Sketch a sample profile plot of the sample male mean vector and the sample female mean vector. Does it appear that the null hypothesis is false?

Code

```
> matplot(cbind(ybar1,ybar2), ylab="mean",
+ xlab="variable", main="Profile plot", type="b", lwd=3)
```

- How can you write the null hypothesis in terms of $H_0 : \mathbf{C}\boldsymbol{\mu}_1 = \mathbf{C}\boldsymbol{\mu}_2$? The trick is to recognize that the null hypothesis corresponds to the slopes of each line in the profile plot being parallel. For example: $\mu_{12} - \mu_{11} = \mu_{22} - \mu_{21}$, and $\mu_{13} - \mu_{12} = \mu_{23} - \mu_{22}$, etc.
- Note that you could view testing this hypothesis as testing $H_0 : \boldsymbol{\mu}_{z_1} = \boldsymbol{\mu}_{z_2}$ where $\mathbf{z}_1 = \mathbf{C}\mathbf{y}_1$ and $\mathbf{z}_2 = \mathbf{C}\mathbf{y}_2$.
- What is the test statistic in terms of transformed data?
- What is the test statistic in terms of the original **Y** data?
- What value for p and v should you use to look up the critical value?

- Formally state and test H_0. Interpret the results.

```
Code
> C = matrix(c(1,-1,0,0, 0,1,-1,0, 0,0,1,-1), byrow = T, nrow=3)
> z1 = as.matrix(males)%*%t(C)
> z2 = as.matrix(females)%*%t(C)
> print(hotelling.test(z1,z2))
Test stat:  74.24
Numerator df:  3
Denominator df:  60
P-value:  2.587e-10
```

5.5 Permutation tests

Example 5.6. *Let us revisit the psychological data where recall 4 exams were given to males and 4 exams were given to females.*

- Based on the biplot for males and the biplot for females, do you think normality is questionable?
- What is the distribution of T^2 under the null hypotheses $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ when normality is not satisfied?

Permutation tests fall under the category of *nonparametric tests* in the sense that they do not require that a parametric model, such as a normal model, for the data be satisfied. The idea is to estimate the distribution of the test statistic under the null hypothesis using the data rather than deriving it under a parametric model.

1. The first step is to estimate the distribution of the test statistic under the null hypothesis.

- First, take the 64 measurement vectors and randomly assign "male" to 32 of them and "female" to the other 32 (permute) and compute a test statistic, say T_1^2.
- Repeat this process 10,000 times to get $T_1^2, T_2^2, ..., T_{10000}^2$. These permuted test statistics constitute a sample from the distribution of T^2 under the assumption that the distribution of male test scores is equal to the distribution of female test scores.

2. Next we compare the original test statistic to the estimated distribution of T^2 under the null hypotheses.

   - Recall that the p-value is the probability of getting data more extreme than what you got assuming the null hypothesis is true. If you take a sample T^2 from the population (permuted), what is the probability that it will be greater than $T^2 = 23.22$ if the null hypothesis is true? What is the p-value?

Code
```
> plot(hotelling.test(males,females, perm = T))
```

- Formally state the null hypothesis and your conclusions.

Code
```
> print(hotelling.test(males,females, perm = T))
stat:  23.22
Numerator df:  4
Denominator df:  59
Permutation P-value:  0
Number of permutations : 10000
```

Comments:

- Permutation tests can be constructed in a variety of settings. The challenge is knowing how to "permute".
- The advantage of permutation tests is that they are valid regardless of the parametric model for the data.
- The weaknesses of permutation tests
  - They are not always applicable. Could you utilize a permutation test for the calcium/farm null hypotheses?
  - They are computationally intense and sometimes not available in software.
    - Basic large sample theory implies that if you do enough *random* permutations (usually 10,000), then your p-value approximation is accurate. The standard error is less than $0.5/\sqrt{10,000}$.
    - If you don't want to rely on this approximation, you can consider all possible permutations? How many are there for the psychological data? Most software uses an approximated p-value.
  - They fail in extremely small sample size settings, which is when they are needed the most.
    - Suppose that you had $n_1 = n_2 = 3$. How many permutations are there? What is the smallest possible p-value?
  - They assume that data from populations are identical under the null hypothesis.
    - Is it possible that $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ is true in the psychological data but that the populations are not identical?
    - How should you interpret a rejected null hypothesis if using a permutation test?

6 Multivariate analysis of variance (MANOVA)

Outline

- One-way ANOVA (review)
- One-way MANOVA
- Four MANOVA tests
- Measures of multivariate association
- Profile analysis
- Contrasts
- Two-way MANOVA
- Testing additional items
- Variable selection
- Exact test
- Tests for covariance structure

Example 6.1. *Eight apple trees from each of 6 different rootstocks were sampled. A rootstock is a plant with a healthy root system and is used for grafting a cutting or budding from another plant. The variables are*

$y_1 =$ *trunk girth at 4 years (hundreds of mm)*
$y_2 =$ *extension growth at 4 years (m)*
$y_3 =$ *trunk girth at 15 years (hundreds of mm)*
$y_4 =$ *weight of tree above ground at 15 years (thousands of lbs).*

- How many populations are there?
- Do you think that the size of a tree depends on the rootstock from which it was grafted?

Rootstock	Y1	Y2	Y3	Y4
1	1.11	2.57	3.58	0.76
1	1.19	2.93	3.75	0.82
1	1.09	2.87	3.93	0.93
1	1.25	3.84	3.94	1.01
1	1.11	3.03	3.60	0.77
1	1.08	2.34	3.51	0.73
1	1.11	3.21	3.98	1.21
1	1.16	3.04	3.62	0.75
2	1.05	2.07	4.09	1.04
2	1.17	2.88	4.06	1.09
2	1.11	3.38	4.87	1.64
2	1.25	3.91	4.98	1.52
2	1.17	2.78	4.38	1.20
2	1.15	3.02	4.65	1.24
2	1.17	3.38	4.69	1.50
2	1.19	3.45	4.40	1.03
⋮	⋮	⋮	⋮	⋮
6	1.11	2.81	3.76	0.80
6	0.75	0.84	3.14	0.61
6	1.05	2.20	3.75	0.79
6	1.02	2.13	3.99	0.85
6	1.05	1.95	3.34	0.61
6	1.07	2.25	3.21	0.56
6	1.13	3.06	3.63	0.71
6	1.11	2.47	3.95	0.95

Table 2: Size measurements of 8 different apple trees from 6 different root stocks.

```
Code
> root = read.table('Data/T6_2_ROOT.dat', col.names=
+                     c("Rootstock","Y1", "Y2", "Y3", "Y4"))
```

6.1 One-way ANOVA

6.1.1 Data

Sample 1 from $N(\mu_1, \sigma^2)$	Sample 2 $N(\mu_2, \sigma^2)$	... ...	Sample k $N(\mu_k, \sigma^2)$
y_{11}			
y_{12}			
$\vdots$			
y_{1n}			
$y_{1\cdot}$			
$\bar{y}_{1\cdot}$			
s_1^2			

- Describe y_{ij}

6.1.2 Model

- Assumptions:
  - $y_{ij} = \mu + \alpha_i + \epsilon_{ij} = \mu_i + \epsilon_{ij}$

- $\epsilon_{ij} \overset{i.i.d.}{\sim} N(0, \sigma^2)$

- Terminology:
  - What are μ_i, μ, and α_i?

$H_0 : \mu_1 = \mu_2 = \ldots = \mu_k$ vs. H_1 : not H_0

6.1.3 Test

The F statistic depends upon two estimates of σ^2

- The *within* sample estimate is

$$s_e^2 = \frac{1}{k}\sum_{i=1}^{k} s_i^2 = \frac{\sum_{ij}(y_{ij} - \bar{y}_{i\cdot})^2}{k(n-1)} = \frac{SSE}{k(n-1)} = MSE.$$

  - Under H_0 or H_1, $SSE/\sigma^2 \sim \chi^2_{k(n-1)}$ and $E[MSE] = \sigma^2$.

- The *across* sample estimate

$$ns_{\bar{y}}^2 = \frac{n\sum_{i=1}^{k}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}{k-1} = \frac{SSH}{k-1} = MSH$$

  - Under H_0, $SSH/\sigma^2 \sim \chi^2_{k-1}$ and $E[MSH] = \sigma^2$
  - Under H_1, $SSH/\sigma^2 \sim \chi^2_{k-1}$ has a noncentral χ^2 distribution and

$$E[MSH] = \sigma^2 + n\sum_i \alpha_i^2/(k-1) > \sigma^2$$

- We typically construct an ANOVA table

Source	Sums of Squares	df	Mean Squared	F
Model(Hypothesis)				
Error				
Total				

- Under H_0, what is the distribution of F?
- Why would large values of F be evidence against H_0?
- What is an α level test procedure?

6.2 One-way MANOVA

6.2.1 Data

Sample 1 from $N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$	Sample 2 $N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$	... ...	Sample k $N_p(\boldsymbol{\mu}_p, \boldsymbol{\Sigma})$
$\mathbf{y}_{11}$			
$\mathbf{y}_{12}$			
$\vdots$			
$\mathbf{y}_{1n}$			
$\mathbf{y}_{1\cdot}$			
$\bar{\mathbf{y}}_{1\cdot}$			

- Describe y_{ijr}, $\mathbf{y}_{ij}$ and $\bar{\mathbf{y}}_{i\cdot}$

6.2.2 Model

- Assumptions
  - $\mathbf{y}_{ij} = \boldsymbol{\mu} + \boldsymbol{\alpha}_i + \boldsymbol{\epsilon}_{ij} = \boldsymbol{\mu}_i + \boldsymbol{\epsilon}_{ij}$
  - $\boldsymbol{\epsilon_{ij}} \overset{i.i.d.}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$.
- Terminology

- What are $\boldsymbol{\mu}_i$, $\boldsymbol{\mu}$ and $\boldsymbol{\alpha}_i$?

- Hypothesis: $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 = \ldots = \boldsymbol{\mu}_k$ vs. H_1 : not H_0

6.2.3 Tools for Tests: Within vs. across sums of squares

- $\mathbf{E} = \sum_{ij}(\mathbf{y}_{ij} - \bar{\mathbf{y}}_{i\cdot})(\mathbf{y}_{ij} - \bar{\mathbf{y}}_{i\cdot})'$

$$= \begin{pmatrix} SSE_{11} & SPE_{12} & \ldots & SPE_{1p} \\ SPE_{21} & SSE_{22} & \ldots & SPE_{2p} \\ \vdots & \vdots & \ldots & \vdots \\ SPE_{p1} & SPE_{p2} & \ldots & SSE_{pp} \end{pmatrix}$$

  - What is SSE_{11}? How is it related to an ANOVA?
  - What is SPE_{12}? How is this statistic different from any summary statistic in an ANOVA?
  - The error degrees of freedom in this one-way case is $v_E = k(n-1)$. Is this different from an ANOVA?
  - The pooled sample covariance is $\mathbf{S}_p = \mathbf{E}/v_E$ and provided that $\mathbf{E}$ is of full rank, $E[\mathbf{E}/v_E] = E[\mathbf{S}_p] = \boldsymbol{\Sigma}$ under H_0 or H_1. How does this $\mathbf{S}_p$ compare to the pooled sample covariance from Hotelling's T^2 statistic?
  - In general, $rank(\mathbf{E}) = \min(p, v_E)$. We assume $\mathbf{E}^{-1}$ exist for all tests considered here. This means we need $v_E = k(n-1) \geq p$.

- $\mathbf{H} = n \sum_i (\bar{\mathbf{y}}_{i\cdot} - \bar{\mathbf{y}}_{\cdot\cdot})(\bar{\mathbf{y}}_{i\cdot} - \bar{\mathbf{y}}_{\cdot\cdot})'$

$$= \begin{pmatrix} SSH_{11} & SPH_{12} & \dots & SPH_{1p} \\ SPH_{21} & SSH_{22} & \dots & SPH_{2p} \\ \vdots & \vdots & \dots & \vdots \\ SPH_{p1} & SPH_{p2} & \dots & SSH_{pp} \end{pmatrix}$$

  - What is SSH_{11}? How is it related to a statistic from an ANOVA?
  - What is SPH_{12}? How is this statistic different from any ANOVA statistic?
  - The hypothesis or model degrees of freedom in this one-way case is $v_H = k - 1$. Is this different from the usual ANOVA?
  - In general $rank(\mathbf{H}) = \min(p, v_H) = s$ where v_H is the "hypothesis" or "model" degrees of freedom from the usual ANOVA. When will $\mathbf{H}$ be of full rank, i.e. have rank p?

- Relation to an ANOVA table:

Source	Sums of Squares	df	Mean Squared	F?
Model(Hypothesis)				
Error				
Total				

  - Is the F ratio/statistic well defined?
  - The four main MANOVA tests are be written in terms of the eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$, denoted $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_s$. Why are there only s non-zero eigenvalues?
  - The additional parameters to consider when performing a MANOVA as opposed to an ANOVA are p and $s = \min(p, v_H) = rank(\mathbf{E}^{-1}\mathbf{H})$.

6.3 Four MANOVA Tests

We will consider four MANOVA tests and apply them to the root-stalk data.

Code

```
> fit = manova(cbind(Y1,Y2,Y3,Y4)~Rootstock,data=root)

> library(HoRM)
> SS = SSCP.fn(fit)
> SS

> biplot(princomp(fit$residuals))

> summary(fit,test="Wilks")
> summary(fit,test="Roy")
> summary(fit,test="Pillai")
> summary(fit,test="Hotelling-Lawley")
```

- State the model and overall MANOVA null hypothesis for the rootstock data
- Assess normality using the first 2 PC scores of the residuals $\hat{\boldsymbol{\epsilon}}_{ij} = \mathbf{y}_{ij} - \hat{\boldsymbol{\mu}}_i$ and determine why this approach is reasonable
- What are the eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$?
- Will large or small λ_i be evidence against the null hypothesis? Why?

6.3.1 Wilk's Λ

Wilk's Λ test is based on the **likelihood ratio statistic**

$$\Lambda = \frac{|\mathbf{E}|}{|\mathbf{E}+\mathbf{H}|} = \prod_{i=1}^{s} \frac{1}{1+\lambda_i}$$

- What values of Λ are evidence against the MANOVA null hypothesis?
- The distribution of Λ under H_0 is denoted Λ_{p,v_H,v_E} and lower-tailed critical values $\Lambda_{\alpha,p,v_H,v_E}$ are tabulated in Table A.9. Test the overall null hypothesis at the 0.05 level.

- We can transform Λ to an F statistic to get a p-value. See page 176. It should be pointed out that the transformation need not yield exactly an F distribution in some situations. Does the p-value agree with your above conclusion?

6.3.2 Roy's Largest Root

Roy's largest root test statistic is

$$\theta = \frac{\lambda_1}{1 + \lambda_1}$$

- What values of θ are evidence against H_0?
- The distribution of the θ under H_0 is denoted by $\theta_{s,m,N}$ for $s = \min(v_H, p)$, $m = .5(|v_H - p| - 1)$, and $N = .5(v_E - p - 1)$. Critical values $\theta_{\alpha,s,m,N}$ are in Table A.10.
- θ can be tweeked to have a distribution that is bounded above by an F distribution, thus resulting in a **conservative** hypothesis test. Test the MANOVA rootstock hypothesis using a p-value for Roy's largest root statistic.

6.3.3 Pillai Statistic

Pillai's statistic extends Roy's largest Root statistic. It is

$$V^{(s)} = tr((\mathbf{E} + \mathbf{H})^{-1}\mathbf{H}) = \sum_{i=1}^{s} \frac{\lambda_i}{1 + \lambda_i}$$

- What values of $V^{(s)}$ are evidence against H_0?
- Upper critical values for $V^{(s)}_{\alpha,s,m,N}$ are in Table A.11.
- $V^{(s)}$ can be tweaked to have a distribution that is approximated by an F distribution under H_0. Test the MANOVA rootstock hypothesis using a p-value for Pillai's statistic.

6.3.4 Hotelling's Generalized T^2

Hotelling's T^2 test statistic is

$$U^{(s)} = tr(\mathbf{E}^{-1}\mathbf{H}) = \sum_{i=1}^{s} \lambda_i$$

- What values of $U^{(s)}$ are evidence against H_0?
- Upper critical values for $\frac{v_E}{v_H}U^{(s)}$ are in Table A.12.
- We can tweak this statistic to have a distribution that is approximated by an F distribution under H_0. Test the MANOVA rootstock hypothesis using a p-value for Hotelling's test.

6.3.5 Comparison of Tests

- All tests except for Roy's are robust with respect to violations of normality especially when v_E is large. Roy's test is particularly sensitive to outliers. Is Roy's test reasonable here? Why?
- All tests except for Roy's are robust with respect to violations of the equal covariance assumption, when $n_1 \approx n_2 \approx ... \approx n_k$ and each n_i is large. We will consider tests for this equal covariance assumption in the next section. Is it important for this data?
- Roy's test is the most powerful when data are highly correlated, and Wilk's test is the least powerful. When data are nearly uncorrelated, the opposite is true. Based on the PCA of the residuals, which do you think will be the most powerful for the rootstock data?

6.4 Measures of multivariate association

Though we have seen that there is a rootstock effect on tree size, we may wish to know how much of an effect exists. Recall that the standard measure of

association in the usual ANOVA is

$$\eta^2 = \frac{\text{between sum of squares}}{\text{total sums of squares}}$$

For multivariate association,

1. Wilks Λ can be used
$$\eta_\Lambda^2 = 1 - \Lambda = \frac{|\mathbf{H}|}{|\mathbf{E}+\mathbf{H}|}$$

2. We can view Roy's root
$$\eta_\theta^2 = \theta = \frac{\lambda_1}{1+\lambda_1} = \frac{SSH(z)}{SSE(z)+SSH(z)} = r_1^2.$$
where $z = \mathbf{a}_1'\mathbf{y}$ and $\mathbf{a}_1$ is the eigenvector corresponding to λ_1.
   - It is sometimes called the first squared canonical correlation coefficient.
   - Since $\mathbf{a}_1$ is the first eigenvector of $\mathbf{E}^{-1}\mathbf{H}$, it maximizes $SSH(\mathbf{a}'\mathbf{y})/SSE(\mathbf{a}'\mathbf{y})$ over all $\mathbf{a}$. Hence, it can be thought of as the largest possible attainable R^2 value for an ANOVA model if reducing the data to 1 dimension with a linear transformation. Why would Roy's root always be larger than the R^2 value for any individual variable?

3. We can view Pillai's statistic as the sum of the squared canonical correlations
$$V^{(s)} = \sum_{i=1}^{s} \frac{\lambda_i}{1+\lambda_i} = \sum_{i=1}^{s} r_i^2$$
and compute the average
$$A_P = \frac{\sum_{i=1}^{s} r_i^2}{s} = \frac{V^{(s)}}{s}$$

4. For the generalized Hotelling's $U^{(s)} = \sum_{i=1}^{s} \lambda_i$ we compute
$$A_H = \frac{U^{(s)}/s}{1+U^{(s)}/s}$$

For the rootstock data, you could verify that the measures of association are

Measure	Value
η^2_Λ	.846
η^2_θ	.652
A_P	.326
A_H	.422

- How do you interpret η^2_θ?

In summary,

	Wilks	Roys	Pillai	Hotelling
Test Stat				
Test Stat Eigen vals				
Robust				
Power				
p-value properties				
Measure of Assoc.				

6.5 Profile analysis

- We can consider null hypotheses of the form $H_0 : \mathbf{C}\boldsymbol{\mu}_1 = \mathbf{C}\boldsymbol{\mu}_2 = ... = \mathbf{C}\boldsymbol{\mu}_k$ for some $\mathbf{C}$.
  - Sketch a profile plot under which the null hypothesis that the difference across rootstock populations does not depend on the tree size variable is true.
  - What is $\mathbf{C}$ for this hypothesis? Is it unique?
  - Sometimes this null hypothesis is written H_0 : The k profiles are parallel.
- Just as before, we are essentially transforming our data via $\mathbf{z}_{ij} = \mathbf{C}\mathbf{y}_{ij}$ and testing $H_0 : \boldsymbol{\mu}_{z_1} = \boldsymbol{\mu}_{z_2} = ... = \boldsymbol{\mu}_{z_k}$.
- We can compute $\mathbf{E}_z = \mathbf{CEC}'$ and $\mathbf{H}_z = \mathbf{CHC}'$ and get any of the four test statistics using $\mathbf{E}_z$ and $\mathbf{H}_z$.
- As before, we use $p = rank(\mathbf{C})$.

Example

Code

```
> c = matrix(c(1, 0, -1, 0), nrow=1, byrow = T)
> z = as.matrix(root[,-1])%*%t(c)
> profile1 = data.frame(root,z)
> anova(lm(z~Rootstock,data=profile1))
```

- Consider the hypothesis $H_0 : \mathbf{c}\boldsymbol{\mu}_1 = \mathbf{c}\boldsymbol{\mu}_2 = ... = \mathbf{c}\boldsymbol{\mu}_6$ for $\mathbf{c} = (1, 0, -1, 0)$.
- Describe this null hypothesis in words and interpret the p-value.

```
Code
> C = matrix(c(1, -1, 0, 0, 1, 0, -1, 0, 1, 0, 0, -1),
+            nrow=3, byrow = T)
> Z = as.matrix(root[,-1])%*%t(C)
> profile2 = data.frame(root,Z)
> names(profile2)[6:8] = c("Z1","Z2","Z3")
> summary(manova(cbind(Z1,Z2,Z3)~Rootstock,data=profile2))
```

- Formally define the null hypothesis.
- Interpret the p-value.

6.6 Contrasts

- We may wish to test $H_0 : \boldsymbol{\delta} = 0$ against $H_1 : \boldsymbol{\delta} \neq 0$ where

$$\boldsymbol{\delta} = c_1\boldsymbol{\mu}_1 + c_2\boldsymbol{\mu}_2 + ... + c_k\boldsymbol{\mu}_k$$

  and the where the c_i's sum to 0.

- Again, we are essentially transforming our data via $\hat{\boldsymbol{\delta}} = \sum_{i=1}^{k} c_i\bar{\mathbf{y}}_{i\cdot}$ and hence we can make use of the transformation theorem and compute Hotelling's T^2 test stat via $T^2 = \hat{\boldsymbol{\delta}}'[\widehat{Cov(\hat{\boldsymbol{\delta}})}]^{-1}\hat{\boldsymbol{\delta}}$. What is T^2 in terms of the original $\mathbf{S}_p$ and $\bar{\mathbf{y}}_{i\cdot}$, c_i, and n?

- We can also get a new $\mathbf{H}_{\hat{\boldsymbol{\delta}}}$ and $\mathbf{E}_{\hat{\boldsymbol{\delta}}}$ matrix and get our test statistics and/or p-values as before. Recall that for we had $s = rank(\mathbf{H}) = \min(v_H, p)$ nonzero eigenvalues. We now have $rank(\mathbf{H}_{\hat{\boldsymbol{\delta}}}) = 1$.

- How many nonzero eigenvalues of $\mathbf{E}_{\hat{\boldsymbol{\delta}}}^{-1}\mathbf{H}_{\hat{\boldsymbol{\delta}}}$ are there? Why will the four MANOVA tests always be identical?

For the rootstock data, we may wish to test the null hypothesis

$$H_0 : \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 = 0$$

Code

```
> root12 = root[root$Rootstock<=2,]
> library(tidyverse)
> root12 = root%>%filter(Rootstock %in% c(1,2))
> library(Hotelling)
> print(hotelling.test(Y1 + Y2 + Y3 + Y4~Rootstock, data = root12))
```

- What is $\boldsymbol{\delta}$ and H_0?
- Interpret the p-value.

Here, a profile analysis is run on the first two populations, i.e., we have a profile analysis and contrast.

Code

```
> C = matrix(c(1,-1,0,0, 0,1,-1,0, 0,0,1,-1), byrow = T, nrow=3)
> z1 = as.matrix(root12[root12$Rootstock==1,-1])%*%t(C)
> z2 = as.matrix(root12[root12$Rootstock==2,-1])%*%t(C)
> print(hotelling.test(z1,z2))
```

- Formally define the null hypothesis
- Interpret the p-value.

6.7 Two-way MANOVA

Example 6.2. *A steel study was a 2×4 design with 4 replications for a total of 32 observational vectors and two variables. Factor A was rotational velocity ($A_1 =$ high and $A_2 =$low) and the four levels of factor B are lubricant type. Variables measured are $y_1 =$ ultimate torque and $y_2 =$ ultimate strain.*

- The model is
$$\mathbf{y}_{ijk} = \boldsymbol{\mu} + \boldsymbol{\alpha}_i + \boldsymbol{\beta}_j + \boldsymbol{\gamma}_{ij} + \boldsymbol{\epsilon}_{ijk}$$
where
  - $\boldsymbol{\alpha}_i$ is the effect of the i'th rotational velocity $i = 1, a = 2$
  - $\boldsymbol{\beta}_j$ is the effect of the j'th lubricant $j = 1, 2, 3, b = 4$
  - $\boldsymbol{\gamma}_{ij}$ is the interaction effect of A_i and B_j.
  - Here, we have the k'th experimental unit in the ij'th treatment group $k = 1, 2, 3, n = 4$.
  - $\boldsymbol{\epsilon}_{ijk} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$
- The MANOVA table is

Source	Sums of Squares	df
$\mathbf{H}_A$	$= nb \sum_i (\bar{\mathbf{y}}_{i..} - \bar{\mathbf{y}}_{...})(\bar{\mathbf{y}}_{i..} - \bar{\mathbf{y}}_{...})'$	$a - 1$
$\mathbf{H}_B$	$=$ analogous	$b - 1$
$\mathbf{H}_{AB}$	$=$ analagous	$(a-1)(b-1)$
$\mathbf{E}$	$= \sum_{ijk} (\mathbf{y}_{ijk} - \bar{\mathbf{y}}_{ij.})(\mathbf{y}_{ijk} - \bar{\mathbf{y}}_{ij.})'$	$ab(n-1)$

- We can perform a hypothesis test by getting the eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$.
- How would you test the null hypothesis that there is no lubricant-velocity interaction?
- How would you test the null hypothesis that there is no lubricant effect?

6.8 Testing additional information

Given that we have already measured trunk girth at 4 years (y_1) and extension growth at 4 years (y_2), do trunk girth at 15 years and weight of a tree at 15 years provide any additional information for discriminating between populations?

Partial Wilks Λ:

- The data are
$$\begin{pmatrix} \mathbf{y}_{ij} \\ \mathbf{x}_{ij} \end{pmatrix}$$
  - $i = 1, 2, ..., k$ and $j = 1, 2, ..., n$
  - $\mathbf{y}_{ij}$ is q dimensional and $\mathbf{x}_{ij}$ is $p - q$ dimensional
- We can get
$$\mathbf{E} = \begin{pmatrix} \mathbf{E}_{yy} & \mathbf{E}_{yx} \\ \mathbf{E}_{xy} & \mathbf{E}_{xx} \end{pmatrix} \text{ and } \mathbf{H} =$$
- The usual Wilk's λ statistic is
$$\Lambda(\mathbf{y}, \mathbf{x}) = \frac{|\mathbf{E}|}{|\mathbf{E} + \mathbf{H}|}$$
  which is distributed according to Λ_{p,v_H,v_E} under the null hypothesis.
- We could also compute
$$\Lambda(\mathbf{y}) = \frac{|\mathbf{E}_{yy}|}{|\mathbf{E}_{yy} + \mathbf{H}_{yy}|}$$
  which is distributed according to Λ_{q,v_H,v_E}
- The hypotheses are

  H_0 : Given $\mathbf{y}$, variables in $\mathbf{x}$ do not help discriminate between populations
$$H_1 : \text{ not } H_0$$
- The test statistic is
$$\Lambda(\mathbf{x}|\mathbf{y}) = \frac{\Lambda(\mathbf{x}, \mathbf{y})}{\Lambda(\mathbf{y})}$$

- It turns out that $\Lambda(\mathbf{x}|\mathbf{y}) \sim \Lambda_{p-q,v_H,v_E-q}$.
- Will small or large values of $\Lambda(\mathbf{x}|\mathbf{y})$ be evidence against H_0?

Code

```
> fit = manova(cbind(Y1,Y2,Y3,Y4)~Rootstock,data=root)
> summary(fit,test="Wilks")

> fit2 = manova(cbind(Y1,Y2)~Rootstock,data=root)
> summary(fit2,test="Wilks")

> SS = SSCP.fn(fit)
> D1 = det(SS$SSCPE)/det(SS$SSCPTO)
> M = matrix(c(1,0,0,0, 0,1,0,0), byrow = T, nrow=2)
> D2 = det(M%*%SS$SSCPE%*%t(M))/det(M%*%SS$SSCPTO%*%t(M))

> D1/D2
```

- Observe that $\Lambda(y_1, y_2, y_3, y_4) = 0.154$ and $\Lambda(y_1, y_2) = 0.700$. Hence, $\Lambda(y_3, y_4|y_1, y_2) = \frac{.154}{.700} = .22$,
- What is the critical value for a $\alpha = .05$ level test?
- Given the tree size measurements at 4 years, do the tree size measurements at 15 years help discriminate between populations?

6.9 Variable selection

- How is multivariate variable selection different from variable selection in the univariate setting? Are we trying to select response or explanatory variables?
- Selecting a collection of response variables is sometimes referred to as a **stepwise discriminant analysis** or a **stepwise MANOVA**.

6.9.1 Forward Selection

1. Compute $\Lambda(y_i)$ for each of the y_i univariate variables. Include the y_i with the most significant test in the model, call it $y_{(1)}$.
2. Compute $\Lambda(y_i|y_{(1)})$ for the remaining $p-1$ variables. Include the y_i with the most significant partial Λ in the model if it is significant.
3. Compute $\Lambda(y_i|y_{(1)}, y_{(2)})$ for the remaining $p-2$ variables ...

- Comment: since we are adding 1 variable at a time we may also use the fact that
$$\frac{1-\Lambda_{1,v_H,v_E-q}}{\Lambda_{1,v_H,v_E-q}}\frac{v_E-q}{v_H} = F_{v_H,v_E-q}$$
to get an exact p-value.

Code

```
> n = 8
> k = 6
> v_H = k-1
> v_E = k*(n-1)
> candidate = names(root)[-1]
> # 1. Select the most significant variable
> fit = manova(cbind(Y1,Y2,Y3,Y4)~Rootstock,data=root)
> SS = SSCP.fn(fit)
> ts1 = abs(diag(SS$SSCPE))/abs(diag(SS$SSCPTO))
> select1 = candidate[which.min(ts1)]
> D1 = min(ts1)
> candidate = candidate[-which.min(ts1)]
> # 2. Select the most significant variable + Y4
> q = 1
> M = matrix(c(1,0,0,0, 0,0,0,1), byrow = T, nrow=2)
> D2.1 = det(M%*%SS$SSCPE%*%t(M))/det(M%*%SS$SSCPTO%*%t(M))
> ts.1 = D2.1/D1
> pval.1 = 1-pf((1-ts.1)/ts.1*(v_E-q)/v_H,v_H,v_E-q)
> M = matrix(c(0,1,0,0, 0,0,0,1), byrow = T, nrow=2)
> D2.2 = det(M%*%SS$SSCPE%*%t(M))/det(M%*%SS$SSCPTO%*%t(M))
> ts.2 = D2.2/D1
> pval.2 = 1-pf((1-ts.2)/ts.2*(v_E-q)/v_H,v_H,v_E-q)
> M = matrix(c(0,0,1,0, 0,0,0,1), byrow = T, nrow=2)
> D2.3 = det(M%*%SS$SSCPE%*%t(M))/det(M%*%SS$SSCPTO%*%t(M))
> ts.3 = D2.3/D1
> pval.3 = 1-pf((1-ts.3)/ts.3*(v_E-q)/v_H,v_H,v_E-q)
> if(sum(pval<0.05)==0){
+   select = select1
+ }else{
+   select2 = candidate[which.min(pval)]
+   select = c(select1,select2)
+ }
.
.
.
```

1. $\Lambda(y_4)$ is the smallest and is significant when compared to $\Lambda_{.05,q,v_H,v_E} \approx .766$. Note that $q = 1, v_H = 5, v_E = (8-1)6 \approx 40$.

2. $\Lambda(y_3|y_4) = .265/.409 = .648$ and the corresponding p-value is 0.0026, so

we add y_3 to the model.

3. $\Lambda(y_2|y_3, y_4) = .160/.265 = .604$ and the corresponding p-value is 0.0008, so we add y_2 to the model.

4. Finally, we see that $\Lambda(y_1|y_2, y_3, y_4) = .154/.160 = .963$ and the corresponding p-value is 0.9103, so we do not add y_1 to the model.

- Which variables should we use to discriminate between rootstock populations?

Code

```
> folda(root[,-1], root$Rootstock,
+       subsetMeth="forward", testStat="Wilks", alpha=0.05)
```

- What are the partial F statistics and variables added at each step

6.9.2 Other Selection Methods

- In **stepwise selection**, after another variable is included in the model, the model is re-evaluated to determine if another variable should be removed. It could be the case that another variable is no longer significant conditionally on the rest and should be removed.
  - We could get caught in a loop here.
- In **backwards selection**, we start with all the variables and remove the variable that is least significant (conditionally on the rest).
- Change 'FORWARD' to 'STEPWISE' or 'BACKWARD' in the SAS code to run these.

- Any three of these stepwise methods may lead to a different model.
- Using a level $.05$ cutoff for each partial Λ is arbitrary in that there is no control of an experiment-wise error rate. This choice should reflect your desire to discriminate between populations with a small number of variables.
- Two different MANOVA models will typically result in different stepwise discriminant results.

6.10 Exact test

Code

```
> library(vegan)
> fit_exact = adonis2(root[,-1]~Rootstock,data=root,
+                     permutations=1000)
> print(fit_exact)
```

- An exact permutation test works as before. It permutes the STOCK variable 10000 times and gets a permutation distribution for Λ. It can be computationally intense, but it no longer requires normality. The null hypothesis is that all data are i.i.d.
- Get the permutation p-value and interpret it.
- Could you get a permutation p-value for profile analyses and/or contrasts?

In summary,

- Univariate analysis is called protected if it is performed at level α only when the overall MANOVA null hypothesis is rejected at level α. This

approach should reasonably control the experimentwise error rate if p is small, say less than 5 or 10, but experimentwise error rate control is not guaranteed.

- For most any type of univariate ANOVA design, an extension to MANOVA exists.
- Assumptions: All four tests, except Roy's, are reasonably robust to violations of the normality assumption and equal covariance assumption.
  - The equal covariance assumption is especially not important when the sample sizes from each population are large and approximately equal.
  - Normality assumptions are less important for larger sample sizes. If they are questionable. Use a permutation test.
- Sometimes data are collected over time. Make sure you understand whether you should test for linear, quadratic, etc., trends using a contrast or profile analysis.
- We have focused on the setting when sample sizes are equal for all samples. This need not be the case.
  - What is the one-way MANOVA model in this setting?
  - The forms of the $\mathbf{H}$ and $\mathbf{E}$ matrices are similar, and tests are still computed using the eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$.
  - Degrees of freedom are

$$v_H = k - 1 \text{ and } v_E = \sum_i n_i - k$$

6.11 Tests for covariance structures

We will consider two different types of null hypotheses regarding covariance structure:

1. $H_0 : \mathbf{\Sigma}_1 = \mathbf{\Sigma}_2 = ... = \mathbf{\Sigma}_k$.

2. $H_0 : \mathbf{\Sigma}_{xy} = \mathbf{0}$.

Example 6.3. *Eight apple trees from each of 6 different rootstocks were sampled. The variables are*

$y_1 =$ *trunk girth at 4 years (hundreds of mm)*
$y_2 =$ *extension growth at 4 years (m)*
$y_3 =$ *trunk girth at 15 years (hundreds of mm)*
$y_4 =$ *weight of tree above ground at 15 years (thousands of lbs).*

- Earlier inferences made for this experiment relied on the model (for $i = 1, 2, ..., 6$; $j = 1, 2, ..., 8$)
$$\mathbf{y}_{ij} = \boldsymbol{\mu}_i + \boldsymbol{\epsilon}_{ij}$$
with $\boldsymbol{\epsilon}_{ij} \overset{iid}{\sim} N_4(\mathbf{0}, \mathbf{\Sigma})$.
- Question: Is the assumption that $\mathbf{\Sigma}_1 = \mathbf{\Sigma}_2 = ... = \mathbf{\Sigma}_6$ reasonable?

Box's M test

- The hypotheses are $H_0 : \mathbf{\Sigma}_1 = \mathbf{\Sigma}_2 = ... = \mathbf{\Sigma}_k$ and H_1 : (not H_0)
- The likelihood ratio test statistic is
$$M = \frac{|\mathbf{S}_1|^{v_1/2}|\mathbf{S}_2|^{v_2/2} \cdots |\mathbf{S}_k|^{v_k/2}}{|\mathbf{S}_p|^{\sum_i v_i/2}}$$
  - $v_i = n_i - 1$ for n_i the sample size from population i
  - $\mathbf{S}_i$ is the sample covariance matrix for population i
  - $\mathbf{S}_p = \mathbf{E}/v_E$
- Explanation
  - We can factor the test statistic into $M = \prod_i \left(\frac{|\mathbf{S}_i|}{|\mathbf{S}_p|}\right)^{v_i/2}$
  - Under H_0, should $|\mathbf{S}_i|$ be "close" or "far" from $|\mathbf{S}_p|$?
  - What happens to the test statistic as each $\mathbf{S}_i$ gets further from $\mathbf{S}_p$?

  - Will large or small value of the test statistic be evidence against H_0?

- Critical values for the distribution for $-2logM$, which depends on $v = \sum_i v_i$, k, and p, are in Table 14.
  - Will large or small values of $-2logM$ be evidence against H_0?
  - Box (1949, 1950) showed that the distribution of $-2logM$ is approximately proportional to a χ^2_a with $a = (k-1)p(p+1)/2$ distribution. We can use this result to get a p-value.

Example:

Code
```
> library(heplots)
> boxM(cbind(Y1,Y2,Y3,Y4)~Rootstock,data=root)
```

- What are the null and alternative hypotheses?
- Do you reject the null hypothesis?

Comments

- This test is (an adjusted) likelihood ratio test and hence possesses the usual optimality properties.
- Unlike most MANOVA tests from the previous section, this test is **not robust** to violations of the multivariate normality assumption. Hence, if normality is questionable then we should not rely on this test, but should focus more on $log(|\mathbf{S}_i|)$ vs. $log(|\mathbf{S}_p|)$ and how important it is for $\mathbf{\Sigma}_1 = \mathbf{\Sigma}_2 = ... = \mathbf{\Sigma}_k$ for the particular statistical method at hand.

Example: Reassessing MANOVA assumptions

- Assess the normality assumption for the MANOVA model for the rootstock data using PCA plots of the residuals. Make sure to discuss the robustness of MANOVA tests vs normality and whether or not it's reasonable to use PCA plots to assess normality.

- Assess the equal covariance assumption using Box's M test and by comparing $log(|\mathbf{S}_i|)$ vs. $log(|\mathbf{S}_p|)$. Make sure to comment on whether or not Box's M test is trustworthy and on the robustness of the equal covariance assumption for MANOVA tests.

Tests for Independence of Groups

Example 6.4. *Five measurements were taken on patients with diabetes. Three variables of major interest and two variables of minor interest are*
x_1 = *glucose intolerance*
x_2 = *insulin response to oral glucose*
x_3 = *insulin resistance*
y_1 = *relative weight*
y_2 = *fasting plasma glucose.*

- Are glucose intolerance, insulin response, or insulin resistance related to weight or fasting plasma glucose?

- We have variables $\begin{pmatrix} \mathbf{x}_q \\ \mathbf{y}_p \end{pmatrix}$ with sub mean vectors $\boldsymbol{\mu}_x$ and $\boldsymbol{\mu}_y$, and covariance matrices $\boldsymbol{\Sigma}_{xy}$, $\boldsymbol{\Sigma}_{yx}$, $\boldsymbol{\Sigma}_{xx}$, and $\boldsymbol{\Sigma}_{yy}$.

- We are testing $H_0 : \Sigma_{xy} = \mathbf{0}$ against H_1 :(not H_0).

- The test statistic is
$$\Lambda = \frac{|\mathbf{S}|}{|\mathbf{S}_{xx}||\mathbf{S}_{yy}|}.$$

- Under H_0, $\Lambda \sim \Lambda_{q,p,n-1-q}$.

- To see why small values of Λ are evidence against H_0, consider $p = 1$ and $q = 1$

Example 6.5. *Five measurements were taken on patients with diabetes. Three variables of major interest and two variables of minor interest are*
x_1 = *glucose intolerance*
x_2 = *insulin response to oral glucose*
x_3 = *insulin resistance*
y_1 = *relative weight*
y_2 = *fasting plasma glucose*

- We would like to know if we can learn about the major variables of interest using the minor variables of interest.

Code

```
> diabetes= read.table("T3_5_DIABETES.dat",
col.names=c("Person","Y1","Y2","X1", "X2", "X3"), row.names=1)
> Sy = cov(diabetes[1:2])
> Sx = cov(diabetes[3:5])
> S = cov(diabetes)
> det(S)/(det(Sy)*det(Sx))
[1] 0.7240096
```

- What is the hypothesis test of interest for the diabetes data?
- Are minor variables of interest independent of major variables?

Relation to Canonical Correlation

- The test statistic for independence can be expressed in terms of the eigenvalues of $\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy}\mathbf{S}_{yy}^{-1}\mathbf{S}_{xy}$ given by $r_1^2, r_2^2, ..., r_s^2$. It is

$$\Lambda = \prod_{i=1}^{s}(1 - r_i^2).$$

  - We write r_i^2 since these are also called the squared canonical correlations between $\mathbf{y}$ and $\mathbf{x}$. We will study this more in the later chapter.

- We can come up with Roy's largest root, generalized Hotelling T^2, and Pillai's test for independence using the r_i^2 values.

Tests for covariance structures

- In general, we can test for specific covariance structures via a likelihood ratio test. Some structures are
  - $\boldsymbol{\Sigma} = \sigma^2 \boldsymbol{I}$ (independence)
  - $\boldsymbol{\Sigma} = \sigma^2[(1-\rho)\boldsymbol{I} + \rho \boldsymbol{J}]$ (compound symmetry/ intraclass correlation model)
  - auto-regressive structures

7 Discriminant analysis

Outline

- Motivation
- LDA for two groups
- LDA for k groups
- Number of discriminant functions
- LDA and variable importance

7.1 Motivation

Example 7.1. *Twenty engineer apprentices and twenty pilots were given 6 tests (Travers 1939). The tests were*
$y_1 =$ *IQ*
$y_2 =$ *form relations*
$y_3 =$ *dynamometer*
$y_4 =$ *dotting*
$y_5 =$ *sensory motor coordination*
$y_6 =$ *perservation*

Code

```
> pilot = read.table("T5_6_PILOT.dat")
> names(pilot)<-c("group","IQ","FR","DY","DO","SMC","PE")
> pilot
   group  IQ FR  DY  DO SMC  PE
1      1 121 22  74 223  54 254
2      1 108 30  80 175  40 300
.
.
```

```
.
38     2  97 36 100 252  30 225
39     2 141 37 105 250  27 243
40     2 164 32  76 187  30 264
> app = pilot[1:20,-1]
> pil = pilot[21:40,-1]
> ybar1 = colMeans(app)
> ybar2 = colMeans(pil)
> matplot(cbind(ybar1, ybar2), lty = 3, type = "b",
+ pch = c("A","P"), ylab = "score", xlab = "exam")
> plot(pilot[,-1], pch = pilot[,1], col = pilot[,1])
```

- Three motivating questions
  - In what way are engineers and apprentices different?
  - A job applicant claims that he has completed the apprentice program and is a pilot. But you are uncertain of this. Suppose that you have him take the exams and get his scores. Could you use his exam scores to determine whether or not he is lying?
  - If you determine that he is lying, how could you quantify the uncertainty with your assessment?
- Methods
  - Describing group separation is called **discriminant analysis**. This section deals with discriminant analysis.
  - Classifying a new observation into a group is called **classification analysis**.

7.2 LDA for two groups

- Data: $\mathbf{y}_{ij}$ for $i = 1, 2$ and $j = 1, 2, ..., n_i$ are p-dimensional.

- Transformation approach: To determine if a new unit was a member of population one or two, we could transform $\mathbf{y}_{ij}$ via

$$z_{ij} = \mathbf{a}'\mathbf{y}_{ij}.$$

  and compare the unit's z value to $z_{11}, z_{12}, ..., z_{1n_1}$ to $z_{21}, z_{22}, ..., z_{2n_2}$ by seeing if it was closer to z values of the first sample or the second.

- Choice of **a**: We can choose **a** so as to maximize

$$\frac{(\bar{z}_1 - \bar{z}_2)^2}{s_z^2} = \frac{[\mathbf{a}'(\bar{\mathbf{y}}_1 - \bar{\mathbf{y}}_2)]^2}{\mathbf{a}'\mathbf{S}_p\mathbf{a}} \tag{2}$$

  where s_z is the pooled standard deviation of the z values.

- If $p = 2$, what would a graph of the two populations and **a** look like?

Linear discriminant function:

- It turns out that the optimal choice of **a** in equation (2) is proportional to

$$\mathbf{a} = \mathbf{S}_p^{-1}(\bar{\mathbf{y}}_1 - \bar{\mathbf{y}}_2)$$

- The **discriminant function** is

$$z = \mathbf{a}'\mathbf{y} = a_1y_1 + a_2y_2 + ... + a_py_p$$

  and the a_i's are the coefficients of the linear discriminant function.

Relation to Hotelling's T^2

- Plug **a** into the expression in (2). If Hotelling's T^2 statistic is not statistically significant, what does this tell us about **a**?

Example:

Code

```
> app = pilot[1:20,-1]
> pil = pilot[21:40,-1]
> ybar1 = colMeans(app)
> ybar2 = colMeans(pil)
> Sp = (cov(pil) + cov(app))/2
> a = solve(Sp)%*%(ybar1 - ybar2)
> a
            [,1]
IQ   0.007515209
FR   0.193260187
DY  -0.129182304
DO  -0.042785864
SMC  0.071839177
PE  -0.049062257
> library(MASS)
> analysis = lda(group~IQ + FR +
DY + DO + SMC + PE, + data = pilot)
> coef(analysis)
             LD1
IQ  -0.002910764
FR  -0.074852845
DY   0.050034428
DO   0.016571668
SMC -0.027824493
PE   0.019002618
>plot(analysis)
```

- Write out the linear discriminant function and comment on how well it works based on the plot. Note that **a** is only unique up to a constant.

7.3 LDA for k groups

Example 7.2. *Six head measurements (in cm) were measured on 30 high school football players, 30 college football players, and 30 non-football players. The variables are*

W = width at widest dimension
C = maximum circumference

FB = front to back measurement at eye level
EYE = eye-to-top of head measurement
EAR = ear-to-top-of-head measurement
JAW = jaw width

Code

```
> football = read.table("T8_3_FOOTBALL.dat",
col.names = c("group", "W","C","FB","EYE","EAR","JAW"))
> football
   group    W     C    FB  EYE  EAR  JAW
1      1 13.5 57.15 19.50 12.5 14.0 11.0
2      1 15.5 58.42 21.00 12.0 16.0 12.0
3      1 14.5 55.88 19.00 10.0 13.0 12.0
.
.
.
89     3 15.7 59.00 20.40 12.1 13.0 12.7
90     3 17.3 61.70 20.70 11.9 13.3 13.3
> plot(football[,-1], pch = football[,1], col = football[,1])
```

- Motivating questions
  - How different are the three groups?
  - Are there really 3 groups, or are there 2?

- We can model our data just as we did with one-way MANOVA - though we won't need to assume multivariate normality for discriminant functions since these are just descriptive statistics. Write out the MANOVA model.

- What do the first, second, etc. eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$ and their corresponding eigenvectors tell you?

- Discriminant functions: For $\mathbf{a}_i$ the first, second, ... eigenvectors of $\mathbf{E}^{-1}\mathbf{H}$ the first, second, ... discriminant functions are

$$z_1 = \mathbf{a}_1'\mathbf{y},\ z_2 = \mathbf{a}_2'\mathbf{y},\ \ldots$$

Code

```
> analysis = lda(group ~ W + C + FB + EYE + EAR + JAW,
+  data = football)
> plot(analysis)
> abline(h=0, v = 0, lwd = 3)
> coef(analysis)
             LD1           LD2
W    0.948423100  1.4067750094
C   -0.003639865 -0.0005126312
FB  -0.006439599 -0.0286176430
EYE -0.647483088  0.5402700415
EAR -0.504360916 -0.3839132257
JAW -0.828535064 -1.5288556226
```

- What are the discriminant functions?
- Does it appear that the two discriminant functions separate all three populations, 1 population from the rest, or no populations? How can you use λ_1 and λ_2 to answer this question?

7.4 Number of discriminant functions

We may wish to test if we really have one population, two populations, etc., or equivalently, how many discriminant functions are necessary to distinguish between populations. The procedure makes use of the fact that

$$\Lambda_m = \prod_{i=m}^{s} \frac{1}{1+\lambda_i} \sim \Lambda_{p-m+1,k-m,N-k-m+1}$$

under the MANOVA model.

1. We first test the null hypotheses $H_0 : \lambda_1 = \lambda_2 = ... = \lambda_s = 0$, where $\lambda_1, \lambda_2, ..., \lambda_s$ are the eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$, with Λ_1. If this null hypothesis is rejected, then at least one eigenvalue is nonzero, and hence there are at least two groups, and we continue to step 2. If this null hypothesis is not rejected, what does this mean?

2. The second null hypothesis is $H_0 : \lambda_2 = \lambda_3 = ... = \lambda_s = 0$ and is tested with Λ_2. If it is rejected, we may conclude that there are at least three populations and go on to test to see if there is a fourth. If this null hypothesis is not rejected, what does this mean?

Example:

Code

```
> library(HoRM)
> fit = manova(cbind(W,C,FB,EYE,EAR,JAW)~group, data=football)
> SS = SSCP.fn(fit)
> E = SS$SSCPE; H = SS$SSCPR
> eigens = eigen(solve(E) %*% H)
> as.numeric(eigens$values)
```

- How many *discirminable* populations are there?

7.5 LDA and variable importance

For the pilot data, we get discriminant function

$$z = -0.003IQ - 0.075FR + 0.050DY + 0.017DO - 0.028SMC + 0.019PE$$

- Intuition: Recall that the coefficients correspond to the eigenvector of $\mathbf{E}^{-1}\mathbf{H}$. Intuitively we would think that FR is the most important for discriminating between populations because its coefficient is the largest (in absolute value).
- When should you perform the LDA on the scaled data?

Other ways to measure variable importance:

1. Compare the partial Wilks statistics and their associated partial R^2 values.
2. Compute the correlation between each variable and the discriminant function or equivalently get marginal R^2 values. These correlations are sometimes called loadings in LDA.

Standardized LDA approach

- How does the discriminant function on the standardized variables compare to the nonstandardized variables for the pilot data? Why are they similar/dissimilar?

Partial R-squared

- The partial F statistic is

$$F = \frac{1-\Lambda}{\Lambda}\frac{v_E - p - 1}{v_H}$$

  where

$$\Lambda = \Lambda(y_r | y_1, ..., y_{r-1}, y_{r+1}, ..., y_p).$$

- The partial R-squared is

$$R^2 = 1 - \Lambda$$

- For the pilot data, which variables had the highest partial R^2? How is it interpreted?

- The partial F statistic is

$$F = \frac{1-\Lambda}{\Lambda}\frac{v_E - p - 1}{v_H}$$

where

$$\Lambda = \Lambda(y_r | y_1, ..., y_{r-1}, y_{r+1}, ..., y_p).$$

- The partial R-squared is

$$R^2 = 1 - \Lambda$$

Code

```
library(folda)
> folda(pilot[,-1], pilot$group,
+       subsetMeth="forward", testStat="Wilks", alpha=0.1)
```

- For the pilot data, which variables had the highest partial R^2? How is it interpreted?

Marginal R-squared

- It can be shown that the correlation between each variable and the first linear discriminant function is proportional to the univariate R^2 value. Hence, we may equivalently inspect each marginal R^2 if opting for this route.

Code

```
> cor(predict(analysis)$x,pilot[,-1])
```

- Which variable, when considered by itself, discriminates most between populations?

In summary, we have discussed *several separate topics* in this chapter:

1. What is the transformation of my variables that best separates my k populations?
2. Which variables provide the most information by themselves for discriminating?
3. Which variables provide the most additional information for discriminating?
4. Can I discriminate between all populations? How many populations do I have?

8 Classification analysis

Outline

- Classification with two groups
- Classification for K groups
- Classification error
- Nonparametric methods

Example 8.1. *Four measurements were made on two species of flea beetles (Lubischew, 1962). The two species are Haltica Oleracea (O), for which there were 19 experimental units, and Halitca Carduorum (C), for which there were 20 experimental units. The measurements are*
DT = distance of tansverse groove from posterior border of prothorax (μm)
LE = length of eleytra (in .01 mm)
L2 = length of second antennal joint (μm)
L3 = length of third antennal joint (μm)

Code

```
> beetles = read.table("T5_5_FBEETLES.dat")[-1]
> names(beetles)<-c("Spec","DT","LE","L2","L3")
> beetles
   Spec  DT  LE  L2  L3
1     1 189 245 137 163
2     1 192 260 132 217
3     1 217 276 141 192
.
.
.
38    2 175 271 140 192
39    2 197 303 170 205
```

Our goal is to

1. Define a rule for classifying a new flea beetle as belonging to O or C.
2. Assess the performance of the rule.

8.1 Classification with two groups

8.1.1 LDA classification rule

- To classify a new experimental unit $\mathbf{y}$ into populations 1 or 2, we can use discriminant function $z = \mathbf{a}'\mathbf{y} = (\bar{\mathbf{y}}_2 - \bar{\mathbf{y}}_1)'\mathbf{S}_p^{-1}\mathbf{y}$. Get the linear discriminant function for the beetles data.
- The idea will be to compute $\bar{z}_1 = \mathbf{a}'\bar{\mathbf{y}}_1$ and $\bar{z}_2 = \mathbf{a}'\bar{\mathbf{y}}_2$ and see if the new z is closer to $\bar{z}_1$ or $\bar{z}_2$. If $z > \frac{\bar{z}_1+\bar{z}_2}{2}$, then z is closer to $\bar{z}_2$ and z **is classified into group 2 (G_2)**. For the beetles data, what is the classification rule in terms of $\mathbf{y}$?

Code

```
> library(MASS)
> analysis = lda(Spec~DT + LE + L2 + L3, data = beetles, prior = c(.5,.5)
> coef(analysis)
           LD1
DT -0.09327642
LE  0.03522706
L2  0.02875538
L3  0.03872998
> analysis$means
        DT       LE       L2       L3
1 194.4737 267.0526 137.3684 185.9474
2 179.5500 290.8000 157.2000 209.2500
> analysis$means%*%coef(analysis)
       LD1
1 2.419488
2 6.120841
> mean(analysis$means%*%coef(analysis))
[1] 4.270164
```

- Suppose that a beetle's measurements are $\mathbf{y}' = (181, 255, 146, 183)$. What species is the beetle?

Code
```
> c(181,255,146,183)%*%coef(analysis)
          LD1
[1,] 3.385739
> c(181,255,146,183)%*%coef(analysis) -  4.27
            LD1
[1,] -0.8844253
> newdata = data.frame(DT=181,LE=255,L2=146,L3=183)
> predict(analysis, newdata = newdata)
```

- Are you confident in your classification? How well does the rule work if you classify all of the data? Note that R reports $z_{ij} = \mathbf{a}'\mathbf{y}_{ij} - \frac{\bar{z}_1+\bar{z}_2}{2}$ so that $\mathbf{y}_{ij}$ is classified into group G_1 if $z_{ij} < 0$.

Code
```
> attach(predict(analysis))
> names(predict(analysis))
[1] "class"     "posterior" "x"
> plot(analysis)
> cbind(x, class,beetles)

           LD1 class spec  DT  LE  L2  L3
1  -3.01630526     1    1 189 245 137 163
2  -0.82008677     1    1 192 260 132 217
.
.
.
26  3.15597187     2    2 177 301 166 221
27 -0.20464749     1    2 198 308 141 197
28  1.50151927     2    2 180 286 146 214
.
.
.
38  0.41490371     2    2 175 271 140 192
```

```
39  0.85623942     2    2 197 303 170 205
### This plots the misclassified unit ####
ind = beetles[,1]
ind[27] = 3
plot(beetles[,-1],pch = ind, col=ind)
```

8.1.2 Bayes classification rule

Review:

- Recall Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

  - If B represents data and we are interested in the event A, then given data B we update the probability of event A.
  - Which expressions represent prior probabilities? Which are posterior probabilities?

- Suppose that the density of **y** given **y** is sampled from population G1 is $f(\mathbf{y}|G_1)$ and the density **y** given G2 is $f(\mathbf{y}|G_2)$. Let prior probability of sampling from each population be $p_1 = P(G_1)$ and $p_2 = P(G_2)$. Given **y**, what's the posterior probability of $G1$?

The Bayesian classifier:

- The rule is simple: Classify **y** into G1 if $P(G_1|\mathbf{y}) > 1/2$.
- It can be verified that this rule reduces to: **y is classified into** $G1$ **if**

$$p_1 f(\mathbf{y}|G_1) > p_2 f(\mathbf{y}|G_2)$$

- If using $\hat{f}(\mathbf{y}|G_i) = N_p(\bar{\mathbf{y}}_i, \mathbf{S}_p)$, then the rule reduces to: **classify y into** $G1$ **if**
$$\mathbf{a}'\mathbf{y} < \mathbf{a}'[\bar{\mathbf{y}}_1 + \bar{\mathbf{y}}_2]/2 + log(p_1/p_2)$$
where **a** is the vector of linear discriminant coefficients.

- How does this rule compare to linear discriminant rule if $p_1 = p_2$?

- Get the posterior probabilities for $p_1 = 0.5$

Code

```
## recall p1 = 0.5 above##
cbind(class, beetles$spec, posterior)
   class              1            2
1      1 1 9.999858e-01 1.416942e-05
2      1 1 9.541493e-01 4.585067e-02
3      1 1 9.999950e-01 4.998447e-06
4      1 1 9.994313e-01 5.686925e-04
5      1 1 9.993933e-01 6.066654e-04
6      1 1 9.994393e-01 5.607394e-04
7      1 1 9.999449e-01 5.508401e-05
8      1 1 9.998335e-01 1.664897e-04
9      1 1 8.228425e-01 1.771575e-01
10     1 1 9.990983e-01 9.016827e-04
11     1 1 9.999704e-01 2.959945e-05
12     1 1 9.998905e-01 1.094614e-04
13     1 1 9.997450e-01 2.550024e-04
14     1 1 9.862371e-01 1.376289e-02
15     1 1 9.924818e-01 7.518210e-03
16     1 1 9.904078e-01 9.592242e-03
17     1 1 9.997139e-01 2.861345e-04
18     1 1 9.635109e-01 3.648910e-02
19     1 1 7.825806e-01 2.174194e-01
20     2 2 1.641187e-05 9.999836e-01
21     2 2 1.606631e-01 8.393369e-01
22     2 2 2.573152e-05 9.999743e-01
23     2 2 1.975266e-04 9.998025e-01
24     2 2 5.069782e-05 9.999493e-01
25     2 2 1.918835e-05 9.999808e-01
26     2 2 8.449749e-06 9.999916e-01
27     1 2 6.808047e-01 3.191953e-01
28     2 2 3.842996e-03 9.961570e-01
29     2 2 4.114083e-04 9.995886e-01
30     2 2 2.338514e-06 9.999977e-01
31     2 2 1.991931e-03 9.980081e-01
32     2 2 1.380062e-02 9.861994e-01
33     2 2 1.382913e-05 9.999862e-01
34     2 2 4.985848e-03 9.950142e-01
35     2 2 4.021941e-04 9.995978e-01
36     2 2 4.389336e-01 5.610664e-01
37     2 2 5.533191e-04 9.994467e-01
```

```
38      2 2 1.771605e-01 8.228395e-01
39      2 2 4.033966e-02 9.596603e-01
```

- Suppose that $p_1 = 0.1$. How does the choice of the prior affect the classification performance? Remark: R uses empirical proportions.

Code

```
> analysis2<-lda(Spec~DT + LE + L2 + L3, data = beetles, prior = c(.1,.9)
> attach(predict(analysis2))
> cbind(class, beetles$spec, posterior)
   class              1            2
1      1 1 9.998725e-01 1.275104e-04
2      1 1 6.980872e-01 3.019128e-01
3      1 1 9.999550e-01 4.498423e-05
4      1 1 9.949049e-01 5.095053e-03
5      1 1 9.945664e-01 5.433618e-03
6      1 1 9.949759e-01 5.024117e-03
7      1 1 9.995045e-01 4.955377e-04
8      1 1 9.985036e-01 1.496414e-03
9      2 1 3.404030e-01 6.595970e-01
10     1 1 9.919430e-01 8.057025e-03
11     1 1 9.997337e-01 2.663320e-04
12     1 1 9.990157e-01 9.842906e-04
13     1 1 9.977097e-01 2.290349e-03
14     1 1 8.884194e-01 1.115806e-01
15     1 1 9.361749e-01 6.382509e-02
16     1 1 9.198225e-01 8.017752e-02
17     1 1 9.974307e-01 2.569329e-03
18     1 1 7.458018e-01 2.541982e-01
19     2 1 2.856806e-01 7.143194e-01
20     2 2 1.823567e-06 9.999982e-01
21     2 2 2.082559e-02 9.791744e-01
22     2 2 2.859124e-06 9.999971e-01
23     2 2 2.195126e-05 9.999780e-01
24     2 2 5.633345e-06 9.999944e-01
25     2 2 2.132076e-06 9.999979e-01
26     2 2 9.388681e-07 9.999991e-01
27     2 2 1.915838e-01 8.084162e-01
28     2 2 4.284632e-04 9.995715e-01
```

```
29     2 2 4.572876e-05 9.999543e-01
30     2 2 2.598355e-07 9.999997e-01
31     2 2 2.217182e-04 9.997783e-01
32     2 2 1.552446e-03 9.984476e-01
33     2 2 1.536589e-06 9.999985e-01
34     2 2 5.564492e-04 9.994436e-01
35     2 2 4.470421e-05 9.999553e-01
36     2 2 7.997286e-02 9.200271e-01
37     2 2 6.151015e-05 9.999385e-01
38     2 2 2.336373e-02 9.766363e-01
39     2 2 4.648882e-03 9.953511e-01
```

- Get the posterior probabilities for classifying $\mathbf{y}' = (181, 255, 146, 183)$ using several different priors. Which prior should you use?

Code

```
> analysis1<-lda(Spec~DT + LE + L2 + L3, data = beetles, prior = c(.5,.5))
> analysis2<-lda(Spec~DT + LE + L2 + L3, data = beetles, prior = c(.1,.9))
> analysis3<-lda(Spec~DT + LE + L2 + L3, data = beetles, prior = c(.01,.99))
>
> predict(analysis1, newdata = newdata)
$class
[1] 1
Levels: 1 2

$posterior
          1         2
1 0.9635109 0.0364891

$x
         LD1
1 -0.8844253

> predict(analysis2, newdata = newdata)
$class
[1] 1
Levels: 1 2
```

```
$posterior
          1         2
1 0.7458018 0.2541982

$x
        LD1
1 -2.364966

> predict(analysis3, newdata = newdata)
$class
[1] 2
Levels: 1 2

$posterior
          1         2
1 0.2105606 0.7894394

$x
        LD1
1 -2.698088
```

8.2 Classification for K groups

- A general classisification rule is **assign y to** G_i if $\Pr(G_i|\mathbf{y})$ is larger than $\Pr(G_j|\mathbf{y})$ for all other j.
- Different models for $f(\mathbf{y}|G_i)$ lead to different classification rules. Normal models lead to linear and quadritc rules.
- Plug in Bayes rule to see why $\Pr(G_i|\mathbf{y}) > \Pr(G_j|\mathbf{y})$ if and only if $p_i f(\mathbf{y}|G_i) > p_j f(\mathbf{y}|G_j)$
- For multivariate normal models, we sometimes focus on $log(p_i f(\mathbf{y}|G_i))$

8.2.1 Linear classification rule

- Model: Suppose $f(\mathbf{y}|G_i) \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma})$
- Equal priors: $p_1 = p_2 = ... = p_K$
  - Taking $\hat{f}(\mathbf{y}|G_i) = N(\bar{\mathbf{y}}_i, \mathbf{S}_p)$ we **assign y to** G_i **if** $D_i^2(\mathbf{y})$ **is the smallest where**
  $$D_i^2(\mathbf{y}) = (\mathbf{y} - \bar{\mathbf{y}}_i)'\mathbf{S}_p^{-1}(\mathbf{y} - \bar{\mathbf{y}}_i)$$
  - Observe that to minimize $D_i^2(\mathbf{y})$ wrt i we can maximize the **linear discriminant function**
  $$L_i(\mathbf{y}) = \bar{\mathbf{y}}_i'\mathbf{S}_p^{-1}\mathbf{y} - 0.5\bar{\mathbf{y}}_i'\mathbf{S}_p^{-1}\bar{\mathbf{y}}_i$$
- Unequal priors: $p_i \neq p_j$
  - If we don't assume $p_i = p_j$ then we maximize
  $$L_i(\mathbf{y}) = \bar{\mathbf{y}}_i'\mathbf{S}_p^{-1}\mathbf{y} - 0.5\bar{\mathbf{y}}_i'\mathbf{S}_p^{-1}\bar{\mathbf{y}}_i + log(p_i)$$
  - Is $L_i(\mathbf{y})$ still a linear function of $\mathbf{y}$?

Example 8.2. *Six head measurements (in cm) were measured on 30 high school football players, college football players, and non-football players. The variables are*
W = width at widest dimension
C = maximum circumference
FB = front to back measurement at eye level
EYE = eye-to-top of head measurement
EAR = ear-to-top-of-head measurement
JAW = jaw width

Code
```
football = read.table("T8_3_FOOTBALL.dat",col.names = c("group", "W","C","FB",
football
analysis = lda(group ~ W + C + FB + EYE + EAR + JAW,
```

```
 data = football)
plot(analysis)
abline(h=0, v = 0, lwd = 3)
coef(analysis)
```

- What are $L_1(\mathbf{y})$, $L_2(\mathbf{y})$, and $L_3(\mathbf{y})$ assuming equal priors?
- What is the distance between each sample mean? Is this supported by the plots of the $\mathbf{z}_{ij}$s?

8.2.2 Quadratic classification rule

- Model: Suppose $f(\mathbf{y}|G_i) \sim N_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$
  - Taking $\hat{f}(\mathbf{y}|G_i) = N(\bar{\mathbf{y}}_i, \mathbf{S}_i)$ observe that the distance doesn't simplify to a linear function. Its a quadratic function of $\mathbf{y}$:

$$D_i^2(\mathbf{y}) = (\mathbf{y} - \bar{\mathbf{y}}_i)'\mathbf{S}_i^{-1}(\mathbf{y} - \bar{\mathbf{y}}_i)$$

  - R will assign $\mathbf{y}$ to the G_i for which the generalized distance

$$-2log(p_i\hat{f}(\mathbf{y}|G_i)) = -2log(p_i) + 0.5log(|\mathbf{S}_i|) + (\mathbf{y} - \bar{\mathbf{y}}_i)'\mathbf{S}_i^{-1}(\mathbf{y} - \bar{\mathbf{y}}_i)$$

    is minimized.

Code

```
library(MASS)
analysis2 = qda(group ~ W + C + FB + EYE + EAR + JAW,
 data = football)
analysis2
library(klaR)
football$group = as.factor(football$group)
partimat(group ~ W + C + FB + EYE + EAR + JAW,
         data = football, method="qda")
```

- For the football data and based on Box's M test, should we be using a Quadratic classification rule?

8.2.3 Discussion

- The Bias-Variance Tradeoff:
  - Suppose that all covariances are not equal but we estimate $\mathbf{\Sigma}_i$ with $\mathbf{S}_p$.
    - How does the variability of the estimator compare with the variability of $\mathbf{S}_i$? What about the Bias?
- The Bias-Variance Tradeoff vs. sample size.
  - What happens to the variance of an estimator as n gets large?
  - What happens to the bias of an estimator as n gets large?
  - For large n should you lean towards the more flexible/variable but less biased estimator or lean towards the less flexible/variable but perhaps more biased estimator?
  - What about for small n.
- Bias Variance Tradeoff and modern classification analysis (unsupervised learning)
  - In many modern applications of classification analysis the sample size is extremely large and consequently an extremely flexible (and complex / black box) classification rule is used. We will consider some slightly more flexible classifiers later.

8.3 Classification error

8.3.1 Apparent error rate

- The *apparent error rate* for a classification rule is:

$$\frac{\text{\# missclassified units}}{\text{\# units}}$$

- The *apparent correct classification rate* is:

$$\frac{\text{\# correctly classified units}}{\text{\# units}}$$

- We can look at the apparent error rate and correct classification rate for each group as well.
- For the football data, what are the apparent error and correct classification rates of the quadratic discriminant rule with equal priors? How does this compare to the linear discriminant rule?
- Here, the units that are classified and used to estimate error rates are also used to build the classification function. How do you think this might effect the error rate estimates?

8.3.2 Cross Validation Error Rates

The basic idea for cross validation estimation of error rates is

1. Split the data set into training and test data.
2. Build the classification rule using training data.
3. Apply the classification rule to the test data, and use the result to estimate the error rates.

- The most widely used of these types of error is referred to the *holdout method* or *cross validation*.
  1. For unit 1, we build the classification rule using the remaining data and then classify unit 1.
  2. This process is repeated for units 2, 3, ..., N.
  3. The cross validation error rate is the proportion of misclassified units.
- How do the cross-validation error rates for the football data compare to the apparent error rates?

8.3.3 Comments

- The apparent error rates will be negatively biased. (The estimates are too small on average). Why?
- Cross validation error rates will be positively biased. Why?
- For large sample sizes, since one observation will contribute little to the classification rule, bias is negligible for both methods.

8.4 Nonparametric Methods

8.4.1 Kernel Methods

The rule which classifies $\mathbf{y}$ into the G_i for which $p_i f(\mathbf{y}|G_i)$ is the largest need not estimate $f(\mathbf{y}|G_i)$ with a normal density. We first consider a simple univariate setting.

- Lets suppose we have $y_1, y_2, ..., y_n$ a random sample with unknown density f.
- First, we use the fact that $\Pr(y_0 - h < y < y_0 + h) \approx 2hf(y_0)$, which yields

$$f(y_0) \approx \frac{\Pr(y_0 - h < y < y_0 + h)}{2h}.$$

  Draw a picture of a histogram to better understand this notion.

- We can estimate $\Pr(y_0 - h < y < y_0 + h)$ with the proportion of the data in $(y_0 - h, y_0 + h)$, call it $N(y_0)/n$. This yields

$$\hat{f}(y_0) \approx \frac{\widehat{\Pr(y_0 - h < y < y_0 + h)}}{2h} = \frac{N(y_0)}{2hn}.$$

- We can provide better estimates of $\Pr(y_0 - h < y < y_0 + h)$. To do so, we first express this estimator as

$$\hat{f}(y_0) = \frac{1}{hn} \sum_{i=1}^{n} K\left(\frac{y_0 - y_i}{h}\right)$$

with

$$K\left(\frac{y_0 - y_i}{h}\right) = 1/2 \text{ if } -1 \leq \frac{y_0 - y_i}{h} \leq 1 \text{ and 0 otherwise.}$$

- $K(u)$ is called the *Kernel* and h is called the *Bandwidth*. Density estimates depend upon these two choices.
- This particular kernel is called the boxcar or rectangular or uniform kernel. We could also use the Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}} e^{-.5u^2}$. How do they differ?
- We can use a multivariate Kernal density estimate via

$$\hat{f}(\mathbf{y}_0) = \frac{1}{nh^p} \sum_{i=1}^{n} K\left(\frac{y_{01} - y_{i1}}{h}, \frac{y_{02} - y_{i2}}{h}, ..., \frac{y_{0p} - y_{ip}}{h}\right)$$

- We then get estimated posterior probabilities and classify **assign y to** G_i if $\widehat{\Pr(G_i|\mathbf{y})}$ is the largest.
- For the football data, how does the Gaussian kernel classification error compare to the other classification error.

Code

```
library(ks)
x = as.matrix(football[,-1])
group = as.numeric(football$group)
analysis = kda(x[,1:3], group)
library(plot3D)
plot(analysis,drawpoints=TRUE)
```

- The choice of bandwidth is crucial in KDE and significantly impacts the accuracy of the density estimate and, consequently, the classification performance. The ks package offers data-driven bandwidth selection methods.

8.4.2 K-nearest neighbor

- This method makes use of distance between $\mathbf{y}$ and all other observation vectors
$$(\mathbf{y} - \mathbf{y}_{ij})'\mathbf{S}_p^{-1}(\mathbf{y} - \mathbf{y}_{ij}).$$
It operates as follows:
  1. For a new observation $\mathbf{y}$, compute all N distances from $\mathbf{y}$ to $\mathbf{y}_{ij}$ and rank them from smallest to largest. Retain the K smallest.
  2. **Assign y to G_i if among the K units, more are from group i than any other group**
- Loftsgaarden and Quesenberry (1965) recommend choosing K near $\sqrt{n_i}$.
- For the football data, how well does the K-nearest neighbor method work for $K = 5$.

Code
```
library(class)
knn_predictions <- knn(train = x, test = x,
                       cl = group, k = 5)
library(caret)
confusion = confusionMatrix(knn_predictions,
                            group)
print(confusion)
```

Remarks

- Posterior probabilities should be reported alongside any classification to allow for assessment of the classification error of the classified unit.

- We can use forward, backward, or stepwise model selection rules to eliminate some variables from the classification functions. When will removing variables result in a better classification rule?
- We may wish to first verify that all populations are "discriminable" using the measured variables before deciding on a classification rule.
- Tools for deciding on linear, quadratic, nonparametric rule, or other rules
  - Classification error rates. Be cognizant that error rates can be biased when n_i is small.
  - Normality assessment. Be concerned with the degree of departure vs. n_i.
  - Equal covariance assessment. Be concerned with the degree of departure vs. n_i.
  - In general, always consider the sample size and the bias-variance tradeoff.
- In many modern "big data" type problems, the sample size is in the thousands or millions. Here, we may
  1. Split the data into training data (say 80% of the data) and test data (remaining 20%)
  2. Consider a really flexible "black box" type model and use the training data to define (estimate parameters) for the rule
  3. Estimate the classification error for the rule on the test data
- Many different distances could be used in the K-nearest neighbor method.

9 Multivariate linear regression

Outline

- Univariate linear regression
- Multivariate linear regression
- Least squares estimation

Example 9.1. *Box and Youle (1955) were interested in understanding how the temperature, concentration level, and exposure time affected several different yield variables in a chemical reaction. The response and explanatory variables are*

y_1 = *percentage of unchanged starting material*
y_2 = *percentage converted to the desired product*
y_3 = *percentage of unwanted by-product*

x_1 = *temperature (degrees F)*
x_2 = *concentration (percentage)*
x_3 = *exposure time (seconds?)*

Code

```
> chem = read.table("Data/T10_1_CHEM.dat",col.names = c("Unit", "Punch",
"Pconv","Punwan","Temp","Conc","Time"))
```

Model: The **general linear model** (GLM) is $\mathbf{Y} = \mathbf{XB} + \mathbf{\Xi}$ where

1. $\mathbf{Y}$ is our usual $n \times p$ matrix of response variables

2. $\mathbf{X}$ is a $n \times (q+1)$ matrix of covariates or explanatory variables
3. $\mathbf{B}$ is a $(q+1) \times p$ matrix of regression coefficients
4. $\boldsymbol{\Xi}' = [\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2, \ldots, \boldsymbol{\epsilon}_n]$ and $\boldsymbol{\epsilon}_i \overset{i.i.d}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$

Why the GLM?

- MANOVA, multivariate analysis of covariance (MANCOVA), and multivariate regression are all special cases of the general linear model (they each just have a different design matrix $\mathbf{X}$). Hence, if we can understand the GLM, we have a very large toolbox for data analysis.

9.1 Univariate linear regression

- There are n experimental units.
- We measure a single response variable and q explanatory variables on each unit. Why is multiple regression considered a univariate procedure?
- We could write the model for each unit via

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + ... + \beta_q x_{1q} + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + ... + \beta_q x_{2q} + \epsilon_2 \\ \vdots &= \vdots \\ y_n &= \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + ... + \beta_q x_{nq} + \epsilon_n \end{aligned}$$

  with $\epsilon_1, \epsilon_2, ..., \epsilon_n \overset{i.i.d.}{\sim} N(0, \sigma^2)$

- Could you write this model in matrix notation via $\mathbf{y}_{n\times 1} = \mathbf{X}_{n\times(q+1)}\boldsymbol{\beta}_{(q+1)\times 1} + \boldsymbol{\epsilon}_{n\times 1}$?

9.2 Multivariate linear regression

- There are n experimental units.
- We measure $p > 1$ response variables and q explanatory variables on each unit.
- The responses are now organized as

$$\mathbf{Y} = \begin{pmatrix} \mathbf{y}_1' \\ \mathbf{y}_2' \\ \vdots \\ \mathbf{y}_n' \end{pmatrix}$$

- We may model a single column (the r'th) variable of $\mathbf{Y}$ as before, via

$$\begin{aligned} y_{1r} &= \beta_{0r} + \beta_{1r}x_{11} + \beta_{2r}x_{12} + ... + \beta_{qr}x_{1q} + \epsilon_{1r} \\ y_{2r} &= \beta_{0r} + \beta_{1r}x_{21} + \beta_{2r}x_{22} + ... + \beta_{qr}x_{2q} + \epsilon_{2r} \\ \vdots &= \vdots \\ y_{nr} &= \beta_{0r} + \beta_{1r}x_{n1} + \beta_{2r}x_{n2} + ... + \beta_{qr}x_{nq} + \epsilon_{nr} \end{aligned}$$

  and write this in matrix form as before.
- For short we write $\mathbf{y}_{(r)} = \mathbf{X}\boldsymbol{\beta}_{(r)} + \boldsymbol{\epsilon}_{(r)}$.
- We may go a step further and write

$$\mathbf{Y}_{n\times p} = \mathbf{X}_{n\times(q+1)}\mathbf{B}_{(q+1)\times p} + \boldsymbol{\Xi}_{n\times p}$$

  where

$$\boldsymbol{\Xi} = \begin{pmatrix} \boldsymbol{\epsilon}_1' \\ \boldsymbol{\epsilon}_2' \\ \vdots \\ \boldsymbol{\epsilon}_n' \end{pmatrix}$$

  and $\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2, ..., \boldsymbol{\epsilon}_n \overset{i.i.d.}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$.
- If we write $\mathbf{Y} = (\mathbf{y}_{(1)}, \mathbf{y}_{(2)}, ..., \mathbf{y}_{(p)})$ and do likewise for $\mathbf{B}$ and $\boldsymbol{\Xi}$ we see that we really just have p regression models stacked next to each other.
- What is the model for the chemical data?

9.3 Least squares estimation

9.3.1 Univariate linear regression

- The goal in **least squares estimation** is to choose $[\hat{\beta}_0, \hat{\beta}_1, ..., \hat{\beta}_q]' = \hat{\boldsymbol{\beta}}$ so as to minimize

$$SSE = \sum_i \hat{\epsilon}_i^2 = \sum_i (y_i - \hat{y}_i)^2 = \sum_i (y_i - [\hat{\beta}_0 + \hat{\beta}_1 x_{1i} + ... + \hat{\beta}_q x_{qi}])^2 = (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$$

- It can be verified that $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

9.3.2 Multivariate linear regression

- Rather than minimizing $SSE = \boldsymbol{\epsilon}'\boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$, we now want to choose $\hat{\mathbf{B}}$ so as to "minimize"

$$\mathbf{E} = \hat{\boldsymbol{\Xi}}'\hat{\boldsymbol{\Xi}} = (\mathbf{Y} - \mathbf{X}\hat{\mathbf{B}})'(\mathbf{Y} - \mathbf{X}\hat{\mathbf{B}}).$$

- The **least squares estimators** are

$$\hat{\mathbf{B}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

- How does this compare to the least squares estimators in the univariate multiple regression setting? Write $\mathbf{Y} = (\mathbf{y}_{(1)}, \mathbf{y}_{(2)}, ..., \mathbf{y}_{(p)})$ to find out and get the least square estimates for the chemical data.

9.3.3 Properties

- The following properties are satisfied even when the $\boldsymbol{\epsilon}_i$s are not normally distributed.
  - The least squares estimator minimizes both $tr(\mathbf{E})$ and $|\mathbf{E}|$.
  - $\hat{\mathbf{B}}$ is unbiased meaning that $E[\hat{\mathbf{B}}] = \mathbf{B}$. If we wanted to predict the value of $\mathbf{y}$ for a particular $\mathbf{x}_{(q+1)\times 1}$ via $\hat{\mathbf{y}}' = \mathbf{x}'\hat{\mathbf{B}}$, what does this tell us about our prediction?
  - Among all other unbiased estimators of $\mathbf{B}$, $\hat{\mathbf{B}}$ has the smallest variance. Specifically, they are the best linear unbiased estimators (BLUE).
- $\hat{\mathbf{B}}$ is the maximum likelihood estimator for $\mathbf{B}$ under a multivariate normal model.

9.4 GLM and MANOVA

9.4.1 ANOVA

- Consider the simple setting where you have y_{1j} and y_{2j} for $j = 1, 2, 3$.
- The ANOVA model is $y_{ij} = \mu_i + \epsilon_{ij}$
- Could you find an $\mathbf{X}$ matrix so that $y_{ij} = \mu_i + \epsilon_{ij}$ could be written

$$\begin{pmatrix} y_{11} \\ y_{12} \\ y_{13} \\ y_{21} \\ y_{22} \\ y_{23} \end{pmatrix} = \mathbf{X}_{6\times 2} \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} + \begin{pmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{13} \\ \epsilon_{21} \\ \epsilon_{22} \\ \epsilon_{23} \end{pmatrix}$$

- What is $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$?

9.4.2 MANOVA

- Suppose that we now measure two variables so that we will have y_{ij1} and y_{ij2}. Then we can stack the y_{ijr}'s so that we have $\mathbf{Y}_{n\times 2}$.
- Note that the same $\mathbf{X}$ can be used. The difference is that we now have $\boldsymbol{\Xi}_{6\times 2}$ and $[\boldsymbol{\beta}_{(1)}, \boldsymbol{\beta}_{(2)}]$.

Example 9.2. *MANOVA vs. Regression: Suppose that the goal was to measure the effect of wind exposure on the health of a wheat plant. Plants were divided into three treatment groups where they were exposed to no wind, 10 mph wind, or 20 mph wind for 8 hours per day until harvest. At harvest, the total number of flag leaves y_1 and shoot biomass y_2 were measured for each of 10 plants in each treatment group.*

Code

```
FL  SB   W
4   2.3  0
7   3.2  0
5   3.1  10
.
.
.
12  1.2  20
```

- Write out the MANOVA effects model and the corresponding general linear model. What is the R code for running a MANOVA?
- Write out the regression model and the corresponding general linear model. What is the R code for fitting the regression model?
- Which model could be used to
  - determine if a 10 mph wind has an effect on the size of a plant

  - predicts the number of flag leaves or shoot biomass of a plant that is typically exposed to 15 mph wind
  - determine if there is a linear relationship between shoot biomass and wind speed
  - determine if wind speed is related to the size of a plant

- Could you determine if there is a quadratic relationship between the size of a plant and wind speed using a GLM?

9.4.3 MANCOVA

Example 9.3. *Consider Fisher's IRIS data. The data are composed of Setosa, Virginica, and Versicolor species and variables SL, SW, PL and PW.*

- Suppose you took sampled an iris plant and recorded its species, SL and SW but failed to record PL and PW. Which GLM would you use to predict the plant's petal size measurements?

Code

```
> library(MASS)
> data(iris)
> plot(iris[,1:4], pch = as.numeric(iris[,5]))
> model = lda(Species ~  Sepal.Length +  Sepal.Width + Petal.Length + Pet
> plot(model)
>
> attach(iris)
> model1 = manova(cbind(Sepal.Length,Sepal.Width)~ Species)
> model2 = manova(cbind(Sepal.Length,Sepal.Width)~  Species+Petal.Length
> model3 = manova(cbind(Sepal.Length,Sepal.Width)~Species+Petal.Length +
>
> summary(model1)
           Df  Pillai approx F num Df den Df    Pr(>F)
Species     2 0.94531   65.878      4    294 < 2.2e-16 ***
Residuals 147
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> summary(model2)
```

```
              Df  Pillai approx F num Df den Df    Pr(>F)
Species        2 1.08457   85.895      4    290 < 2.2e-16 ***
Petal.Length   1 0.57270   96.498      2    144 < 2.2e-16 ***
Petal.Width    1 0.15156   12.861      2    144 7.259e-06 ***
Residuals    145
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> summary(model3)
                      Df  Pillai approx F num Df den Df    Pr(>F)
Species                2 1.09309   84.972      4    282 < 2.2e-16 ***
Petal.Length           1 0.58436   98.415      2    140 < 2.2e-16 ***
Petal.Width            1 0.15597   12.936      2    140 6.996e-06 ***
Species:Petal.Length   2 0.03914    1.407      4    282    0.2317
Species:Petal.Width    2 0.03721    1.337      4    282    0.2565
Residuals            141
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> predict(model2, data.frame(Petal.Width = 5, Petal.Length=10, Species =
  Sepal.Length Sepal.Width
1     12.71246    7.712481
```

- In general, Multivariate analysis of covariance (MANCOVA) is used when you have multiple Y variables and continuous X variable(s) from multiple populations. Mathematically, it just means that some columns in $\mathbf{X}$ are 1's and 0's, and some are not.

9.5 Hypothesis testing

- We are typically not interested in testing whether the intercept coefficients are nonzero, so we write

$$\mathbf{B} = \begin{pmatrix} \boldsymbol{\beta}_0' \\ \mathbf{B}_1 \end{pmatrix}$$

and test the hypotheses $H_0 : \mathbf{B}_1 = \mathbf{0}$ vs. $H_1 : \mathbf{B}_1 \neq 0$.

  - For MANOVA $\boldsymbol{\beta}_0 = \boldsymbol{\mu}$ and $\mathbf{B}_1$ represents the effects $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ...

- To get **H** and **E** we factor the total variability of **Y** into residual variability and model variability:

$$\mathbf{Y}'\mathbf{Y} - n\bar{\mathbf{y}}\bar{\mathbf{y}}' = (\mathbf{Y}'\mathbf{Y} - \hat{\mathbf{B}}'\mathbf{X}'\mathbf{Y}) + (\hat{\mathbf{B}}'\mathbf{X}'\mathbf{Y} - n\bar{\mathbf{y}}\bar{\mathbf{y}}') = \mathbf{H} + \mathbf{E}$$

- Then the eigenvalues of $\mathbf{E}^{-1}\mathbf{H}$ can be used to compute any of the same four tests statistics.

- For the MANOVA, we had $v_H = k - 1$. This is the rank of the design matrix when H_0 is false (full model) minus the rank of the design matrix when H_0 is true (reduced model). What is v_H now in terms of q?

- We plug in q for v_H to get parameters for all MANOVA test statistics. For example, Wilks Λ was written $\Lambda_{p,v_H,N-v_H-1}$. We may now use $\Lambda_{p,q,N-q-1}$ when looking up critical values.

- For the chemical data, do we reject $H_0 : \mathbf{B}_1 = 0$?

Code

```
> summary(model2, test = c("Wilks"))
              Df   Wilks approx F num Df den Df    Pr(>F)
Species        2 0.09928  156.513      4    288 < 2.2e-16 ***
Petal.Length   1 0.42730   96.498      2    144 < 2.2e-16 ***
Petal.Width    1 0.84844   12.861      2    144 7.259e-06 ***
Residuals    145
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comments

- The general linear model is $\mathbf{Y} = \mathbf{XB} + \boldsymbol{\Xi}$ where

$$\boldsymbol{\Xi} = \begin{pmatrix} \boldsymbol{\epsilon}_1' \\ \boldsymbol{\epsilon}_2' \\ \vdots \\ \boldsymbol{\epsilon}_n' \end{pmatrix}$$

  and $\boldsymbol{\epsilon_i} \overset{i.i.d}{\sim} N_p(\mathbf{0}, \boldsymbol{\Sigma})$

- The MANOVA, Multivariate Regression, MANCOVA are all just General Linear Models. The main difference between them is that some of the X variables may just be 1's and 0's (dummy variables).

- The same techniques and properties for MANOVA typically apply more generally to the GLM.
  - The equal covariance assumption generalizes to "$\boldsymbol{\Sigma}$ does not depend on $\mathbf{X}$."
  - Multivariate normality can again be verified on the residuals as before and is not important for large n (except for Roys test).
  - We still can perform profile analysis.
  - Stepwise selection of y and/or x variables can be performed.

- In regression and MANOVA x variables are viewed as fixed. If they are not then all four tests are still the same, but now we are analyzing the distribution of $\mathbf{Y}|\mathbf{X} = \mathbf{x}$.
  - Example: The test $H_0 : \mathbf{B}_1 = \mathbf{0}$ is actually equivalent to the test $H_0 : \boldsymbol{\Sigma}_{yx} = \mathbf{0}$ when $\mathbf{X}$ is viewed as random. This is true whether $\mathbf{X}$ is composed of 0's and 1's or if $\mathbf{X}$ is continuous.
  - Canonical correlation (next) focuses on measuring the correlation between x variables and y variables. When x variables are dummy variables (MANOVA), we can still measure the correlation. Recall Roy's largest root in MANOVA is the first squared canonical correlation coefficient.

10 Principal component analysis

Outline

- PCA on a covariance matrix
- Biplot

10.1 PCA on a covariance matrix

10.1.1 Mathematics of PCA

Suppose that we have a matrix of data $\mathbf{Y}$ with covariance matrix $\mathbf{S}$.

- We first consider a spectral decomposition of $\mathbf{S}$ via

$$\mathbf{S} = \mathbf{CDC}'$$

  What are $\mathbf{D}$ and $\mathbf{C}$?

- If we transform $\mathbf{y}$ via $\mathbf{z} = \mathbf{C}'\mathbf{y}$,
  - What is $\mathbf{S}_z$?
  - What is $|\mathbf{S}_z|$ and $tr(\mathbf{S}_z)$?
  - How do $|\mathbf{S}_z|$ and $tr(\mathbf{S}_z)$ compare to $|\mathbf{S}|$ and $tr(\mathbf{S})$?

10.1.2 Analysis / terminology

Remark: Most software performs a PCA on the centered data $\mathbf{y}_i^* = \mathbf{y}_i - \bar{\mathbf{y}}$ or on the centered and scaled data $\mathbf{y}_i^* = diag(\mathbf{S})^{-1/2}(\mathbf{y}_i - \bar{\mathbf{y}})$.

- Recall $Cov(\mathbf{y} - \boldsymbol{\mu}) = \boldsymbol{\Sigma}$
- Recall $Cov(diag(\boldsymbol{\Sigma})^{-1/2}(\mathbf{y} - \boldsymbol{\mu})) = \mathbf{P}$

This section focuses on PCA for centered data, and it is assume that the data are centered throughout.

- The first, second, third, ... **principle components** are $z_1 = \mathbf{c}_1'\mathbf{y}$, $z_2 = \mathbf{c}_2'\mathbf{y}$, $z_3 = \mathbf{c}_3'\mathbf{y}$, ... where $\mathbf{c}_i$ are the (normalized) eigenvectors of $\mathbf{S}$.
- The vectors $\mathbf{c}_i$ are called the **loading vectors**.
- The transformed data $z_{1i} = \mathbf{c}_1'\mathbf{y}_i$, $z_{2i} = \mathbf{c}_2'\mathbf{y}_i$, ... are called **principle component scores**.
- Important: Which principal component has the highest variance? What is $Cov(z_i, z_j)$?

Example 10.1. *The data set contains ramus bone length measurements for 20 males at ages 8.5, 9, 9.5, and 10.*

- Get scatter plots and boxplots of the original data. Do you notice any possible outliers? Are they detected using boxplots?

Code
```
> bone = read.table('T3_7_BONE.dat', row.names=1, col.names=
c("Person","8.5", "9.0", "9.5", "10"))
> plot(bone, name = "Scatter plots")
> boxplot(bone, name = "Box plots")
```

- Now get the PC loadings and scores. Verify the first couple of PC scores.

Code
```
> analysis = princomp(bone)
> names(analysis)
[1] "sdev"     "loadings" "center"   "scale"    "n.obs"    "scores"   "ca
> attach(analysis)
```

	8.5	9.0	9.5	10
1	47.80	48.80	49.00	49.70
2	46.40	47.30	47.70	48.40
3	46.30	46.80	47.80	48.50
4	45.10	45.30	46.10	47.20
5	47.60	48.50	48.90	49.30
6	52.50	53.20	53.30	53.70
7	51.20	53.00	54.30	54.40
8	49.80	50.00	50.30	52.70
9	48.10	50.80	52.30	54.40
10	45.00	47.00	47.30	48.30
11	51.20	51.40	51.60	51.90
12	48.50	49.20	53.00	55.50
13	52.10	52.80	53.70	55.00
14	48.20	48.90	49.30	49.80
15	49.60	50.40	51.20	51.80
16	50.70	51.70	52.70	53.30
17	47.20	47.70	48.40	49.50
18	53.30	54.60	55.10	55.30
19	46.20	47.50	48.10	48.40
20	46.30	47.60	51.30	51.80

```
> loadings

Loadings:
     Comp.1 Comp.2 Comp.3 Comp.4
X8.5 -0.474  0.592  0.548 -0.353
X9.0 -0.492  0.406 -0.473  0.608
X9.5 -0.515 -0.304 -0.514 -0.615
X10  -0.517 -0.627  0.460  0.358
> scores[1:2,]
    Comp.1    Comp.2      Comp.3     Comp.4
1 2.522507 0.7302355 -0.07435367 0.14057794
2 5.266811 0.5032162 -0.06231710 0.05625445
> bone[1,]
  X8.5 X9.0 X9.5  X10
1 47.8 48.8   49 49.7
> center
  X8.5   X9.0   X9.5    X10
```

```
48.655 49.625 50.570 51.445
> (bone[1,]-center)
    X8.5   X9.0  X9.5    X10
1 -0.855 -0.825 -1.57 -1.745
> -.474*-.855 -.492*-.825 -.515*-1.57-.517*-1.745
[1] 2.521885
> sum((bone[1,] - center)*loadings[,1])
[1] 2.522507
> sum((bone[2,] - center)*loadings[,1])
[1] 5.266811
> sum((bone[1,] - center)*loadings[,2])
[1] 0.7302355
```

- Construct a scatter plot of the first two PC scores and boxplots of each set of PC scores.

Code

```
> plot(scores, main = "plot of 2 PC scores")
> abline(h = 0, v = 0)
> boxplot(scores, main = "Boxplots of 4 PC scores")
```

- Do you notice any possible outliers? Are they detected using boxplots?
- Get the standard deviations for the PC scores and for the original variables.

Code

```
> sdev
   Comp.1    Comp.2    Comp.3    Comp.4
4.8787246 1.2869374 0.4556917 0.3287985
> sqrt(diag(cov(bone)))
    X8.5     X9.0     X9.5      X10
```

```
2.515944 2.539555 2.630209 2.726377
> summary(analysis)
Importance of components:
                          Comp.1     Comp.2      Comp.3      Comp.4
Standard deviation     4.8787246 1.28693742 0.455691682 0.328798463
Proportion of Variance 0.9234897 0.06425905 0.008056783 0.004194488
Cumulative Proportion  0.9234897 0.98774873 0.995805512 1.000000000
```

- How does the standard deviation of the first score compare to the standard deviation of the original variables?
- A **scree plot** is useful for determining how much variability is depicted in each PC. It plots the variance (or standard deviation) of each PC score. Get a scree plot for the data.

Code

```
> plot(analysis, main = "Scree plot of bone data")
```

- Other methods for determining how many scores to use in dimension reduction:
  - retain a fixed percentage of the variability (as defined by the trace), say 80 or 90 percent. What percent of the variance is retained in the first one or two PC scores?

Code

```
> cumsum(sdev^2)/sum(sdev^2)
   Comp.1    Comp.2    Comp.3    Comp.4
0.9234897 0.9877487 0.9958055 1.0000000
```

- Compute the mean of the variances of the PCs and retain a PC if its variance is greater than the mean variance. How many components are retained using this approach?

Code

```
> mean(sdev^2)
[1] 6.443481
> sdev^2
    Comp.1     Comp.2     Comp.3     Comp.4
23.8019538  1.6562079  0.2076549  0.1081084
```

- We can inspect the loadings of the first eigenvector to determine which variables are contributing the most to the overall variability.

Code

```
> loadings

Loadings:
     Comp.1 Comp.2 Comp.3 Comp.4
X8.5 -0.474  0.592  0.548 -0.353
X9.0 -0.492  0.406 -0.473  0.608
X9.5 -0.515 -0.304 -0.514 -0.615
X10  -0.517 -0.627  0.460  0.358

               Comp.1 Comp.2 Comp.3 Comp.4
SS loadings      1.00   1.00   1.00   1.00
Proportion Var   0.25   0.25   0.25   0.25
Cumulative Var   0.25   0.50   0.75   1.00
```

- What are the loading vectors for the first two PCs?
- Notice that the loadings for the first PC are similar in magnitude but are slightly increasing for the first PC. Compare these coefficients to the boxplots of the variables to see why.

- In the second set of PC scores, notice that the three outliers seem to be responsible for most of the variability. Relate the nature of the outliers to the loadings for the second PC.

Suppose that you had a bivariate data set 1: (-1,-1), (0,0), (1,1), (10,10), and data set 2: (-1,-1), (0,0), (1,1), (-1,1).

- Sketch a scatter plot for each data set.
- Describe the loading vectors for each PC.
- Describe the boxplots of the first and second PC scores for each data set. Which boxplot would detect the outlier?
- What if the first variable were multiplied by 10? What would happen to the loadings? Now, which variable contains most of the variability? See below for an illustration.

Code

```
> bone[[1]]<-10*bone[[1]]
> plot(bone)
> loadings(princomp(bone))

Loadings:
     Comp.1 Comp.2 Comp.3 Comp.4
X8.5  0.987  0.139
X9.0        -0.169  0.821  0.536
X9.5        -0.601  0.336 -0.720
X10         -0.769 -0.456  0.440

               Comp.1 Comp.2 Comp.3 Comp.4
SS loadings      1.00   1.00   1.00   1.00
Proportion Var   0.25   0.25   0.25   0.25
Cumulative Var   0.25   0.50   0.75   1.00
> bone[[1]]<-bone[[1]]/10
```

10.1.3 PCA on a correlation matrix

When measurements are on the same scale, it may be reasonable to perform the PCA on the centered data - i.e., use the spectral decomposition for the covariance matrix. If not, we may wish to focus on the centered and scaled data - i.e., use the spectral decomposition for the correlation matrix.

- Get the loadings for the scaled and unscaled bone data? Why are they similar?

Code

```
> princomp(bone, cor = T)$loadings

Loadings:
     Comp.1 Comp.2 Comp.3 Comp.4
X8.5 -0.494  0.584  0.563 -0.313
X9.0 -0.507  0.382 -0.525  0.567
X9.5 -0.509 -0.339 -0.455 -0.648
X10  -0.491 -0.631  0.448  0.401

               Comp.1 Comp.2 Comp.3 Comp.4
SS loadings      1.00   1.00   1.00   1.00
Proportion Var   0.25   0.25   0.25   0.25
Cumulative Var   0.25   0.50   0.75   1.00
> princomp(bone, cor = F)$loadings

Loadings:
     Comp.1 Comp.2 Comp.3 Comp.4
X8.5 -0.474  0.592  0.548 -0.353
X9.0 -0.492  0.406 -0.473  0.608
X9.5 -0.515 -0.304 -0.514 -0.615
X10  -0.517 -0.627  0.460  0.358

               Comp.1 Comp.2 Comp.3 Comp.4
SS loadings      1.00   1.00   1.00   1.00
Proportion Var   0.25   0.25   0.25   0.25
Cumulative Var   0.25   0.50   0.75   1.00

> princomp(bone, cor = F)$scale
X8.5 X9.0 X9.5  X10
   1    1    1    1
```

```
> princomp(bone, cor = T)$scale
    X8.5     X9.0     X9.5      X10
2.452239 2.475253 2.563611 2.657344
```

10.2 Biplot

- Biplots extend a plot of the first two principal component scores by adding lines to indicate the relative contribution of each variable. The line for variable j is drawn by plotting a vector of the coefficients in front of y_j below. For example, y_1 is represented by the vector $(\sqrt{\lambda_1}c_{11}, \sqrt{\lambda_2}c_{21})$. Note that we multiply the loadings for each eigen vector $\mathbf{c}_i$ by the standard deviation of its respective set of scores $\sqrt{\lambda_i}$.

$$z_1 = \sqrt{\lambda_1}c_{11}y_1 + \sqrt{\lambda_1}c_{12}y_2 + ... + \sqrt{\lambda_1}c_{1p}y_p$$
$$z_2 = \sqrt{\lambda_2}c_{21}y_1 + \sqrt{\lambda_2}c_{22}y_2 + ... + \sqrt{\lambda_2}c_{2p}y_p$$

- A biplot is related to the **singular value decomposition** of the data matrix $\mathbf{Y}$. For example, if $\mathbf{Y}$ is the centered and scaled data matrix, then the correlation is computed $\mathbf{R} = \frac{1}{n-1}\mathbf{Y}'\mathbf{Y} = \frac{1}{n-1}(\mathbf{UDV})'\mathbf{UDV}$ where $\mathbf{UDV}$ is the singular value decomposition of $\mathbf{Y}$.

- An interpretation of the plot (if the data is well approximated by two dimensions) is
  - Highly correlated variables point in the same or opposite direction, while independent variables form a right angle.
  - Points that are near one another represent similar units.

- Compare the correlations between variables for the bone data to the angles of lines on the biplot.

Code

```
> cor(bone)
          X8.5      X9.0      X9.5       X10
```

```
X8.5 1.0000000 0.9686511 0.8729938 0.8068860
X9.0 0.9686511 1.0000000 0.9212312 0.8529521
X9.5 0.8729938 0.9212312 1.0000000 0.9659379
X10  0.8068860 0.8529521 0.9659379 1.0000000
> biplot(princomp(bone))
```

10.3 Recap

- Since the eigenvectors are orthogonal, then all principal components are uncorrelated.
- The last couple of principal component scores are often called **structural relationship variables**.
  - Suppose that $\lambda_p \approx 0$, then what is $var(\mathbf{c}_p'\mathbf{y})$?
  - Could you then claim that $\mathbf{c}_p'\mathbf{y}_i \approx 0$ for each i then?
- If the original variables are already nearly uncorrelated, then there is little use for PCA. Why?

11 Factor analysis

Outline

- Factor model
- Estimation of loadings and communalities
- Rotation

11.1 Factor model

11.1.1 The Model

- The idea is that we believe each of the p random variables arises as a linear combination of a few unobservable (usually independent) random variables called **factors**. These are sometimes also called **latent** variables.
- Let m be the number of factors for modeling p variables ($m < p$). Assume (note $\mathbf{y}$ is always centered and scaled because factor analysis is concerned with correlation)

$$\mathbf{y}_{p\times 1} - \boldsymbol{\mu}_{p\times 1} = \boldsymbol{\Lambda}_{p\times m}\boldsymbol{f}_{m\times 1} + \boldsymbol{\epsilon}_{p\times 1}$$

  where $\mathbf{f}$ is a vector of random variables called **common factors**, $\boldsymbol{\epsilon}$ is a vector of (random) errors, $\boldsymbol{\Lambda}$ is a matrix of loadings/coefficients satisfying

  1. $Cov(\mathbf{f}) = \boldsymbol{I}$ and $E(\boldsymbol{f}) = 0$.
  2. $Cov(\boldsymbol{\epsilon}) = \boldsymbol{\Psi}$ where $\boldsymbol{\Psi}$ is a diagonal matrix of **specific variances**, each denoted ψ_i^2
  3. $Cov(\boldsymbol{\epsilon}, \boldsymbol{f}) = \mathbf{0}$.

- Questions:
  - For the ith variable: $y_i - \mu_i =?$
  - Are factors assumed to be uncorrelated with one another? What about common factors and errors?

11.1.2 Example

Suppose that we consider $m = 2$ factors for modeling $p = 5$ variables.

- Write out the model in matrix notation.
- Write out the model for the ith variable y_i.
- What is an expression for the variance of y_i? These two parts are called **specific** variance (ψ_i^2) and **communality** variance (h_i^2).
- In matrix notation, if we take the covariance of both sides of the model equation, we get
$$\mathbf{\Sigma} = \mathbf{\Lambda}\mathbf{\Lambda}' + \mathbf{\Psi}$$
so we are factoring the covariance matrix into the covariance portion and the remaining variance portion, and attempting to understand or interpret the covariance portion $\mathbf{\Lambda}$. Note: Factor analysis really arose out of a theorem that say a symmetric and square matrix $\mathbf{\Sigma}$ of rank $m < p$, can always be factored into $\mathbf{\Sigma} = \mathbf{A}'\mathbf{A}$ where $\mathbf{A}$ is a full rank $m \times p$ matrix. To apply this theory in practice we need $\mathbf{\Psi}$.

11.2 Estimation of loadings and communalities

Since we are interested in interpreting and identifying common factors, most analyses will actually be performed on $\mathbf{R}$ rather than $\mathbf{S}$ (as we mentioned). We will discuss three popular estimation methods, but there are many more.

11.2.1 Principal Component Method

1. We begin by performing a spectral decomposition and factoring the inner diagonal matrix via
$$\mathbf{R} = (\mathbf{C}\mathbf{D}^{1/2})(\mathbf{C}\mathbf{D}^{1/2})'$$

2. If $m = p$ (and it won't be), we would call $\hat{\mathbf{\Lambda}} = \mathbf{CD}^{1/2}$. For m = 2, we would use
$$\hat{\mathbf{\Lambda}} = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 \end{bmatrix} \begin{bmatrix} \sqrt{d_{11}} & 0 \\ 0 & \sqrt{d_{22}} \end{bmatrix}$$

3. To get $\mathbf{\Psi}$, we solve (the diagonal elements of) $\mathbf{R} = \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}' + \mathbf{\Psi}$

11.2.2 Principal Factor/Axis Method

1. Given $\hat{\mathbf{\Psi}}$, get the first m eigenvalues and eigenvectors of $\mathbf{R} - \hat{\mathbf{\Psi}}$ to get $\hat{\mathbf{\Lambda}}$.

2. Get a new $\hat{\mathbf{\Psi}}$ by solving $\mathbf{R} = \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}' + \hat{\mathbf{\Psi}}$ and use it in step 1.

- We repeat steps 1 - 2 until loading estimates "stabilize".
- Note: It is possible that we get nonsensical estimates of communality variance $h_i^2 > 1$, which is nonsensical because it implies a specific variance $\psi_i^2 < 0$.

11.2.3 Maximum Likelihood

- We can assume a normal model with $\mathbf{\Sigma} = \boldsymbol{R} = \mathbf{\Lambda}\mathbf{\Lambda}' + \mathbf{\Psi}$ and get maximum likelihood estimates.
- MLEs also require an iterative procedure.
- The MLE method also requires that $\mathbf{R}$ be full rank.

11.2.4 Example

```
Code
bone = read.table('Data/T3_7_BONE.dat', row.names=1, col.names=
                    c("Person","8.5", "9.0", "9.5", "10"))
library(psych)
f1<-principal(cor(bone),2,rotate="varimax")
f2<-fa(cor(bone),2,fm="pa",rotate="varimax")
f3<-fa(cor(bone),2,fm="ml",rotate="varimax")
```

- Write out the estimated factor model in matrix notation.
- Why might we use $m = 2$?
- Identify the loadings for each factor?
- Write out the model for the first variable and get its communality and specific variance.
- What are the communality and specific variances of the other variables?
- Note that the final communality estimate is the sum of all of the squared loadings, which is $tr(\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}')$.
- Which factor accounts for most of the correlation?
- Do the factors have any meaning? Could a rotation yield more meaningful factors?

11.3 Rotation

11.3.1 Why Rotate?

- The ideal setting is that for each variable y_j, we have say $\lambda_{1j} = 1$ and $\lambda_{2j} = \lambda_{3j} = ...\lambda_{mj} = 0$. We would say that the variable has a *simple*

structure, as opposed to a *complex structure*, if it is composed of one (or only a few) factors.

- Factor loadings are not unique. That is, we may rotate our loadings with an orthogonal transformation $\mathbf{\Lambda}^* = \mathbf{\Lambda T}$. Transformations of this nature will be useful for interpretability. Will these new loadings still satisfy the factorization equation for $\mathbf{\Sigma}$?

11.4 Varimax Rotation

- Using the previous example, plot each variables factor scores with f_1 and f_2 on the x and y axis. The idea with varimax rotation is to rotate the axis so that all four points are very near one of the (still perpendicular) axes.

11.5 Oblique Rotation

- The idea, geometrically, is the same as the varimax rotation, but we can draw any m new axis to make all points almost pass through the new axis. The axes need no longer be orthogonal, however. Sketch a plot of the new oblique axes.
- Important: An *oblique rotation* is not actually rotation at all since we transform according to a non-orthogonal matrix. As a result, the transformed factors are allowed to be correlated, and communalities do not need to be preserved.

11.5.1 Which Rotation

Suppose that you want to build a regression model $y = \beta_0 + \beta_1 x_1 + ...\beta_q x_q$. When q is large, some multicollinearity is inevitable and can cause problems with parameter estimation.

- How might factor analysis be useful?
- Why might you avoid the oblique rotation in this setting?

- Is it ok to model your dependent variables as a linear combination of a smaller subset of still dependent variables in exchange for a simpler structure? This is the question to consider when choosing a rotation.

Code

```
f3<-fa(cor(bone),2,fm="ml",rotate="varimax")
print(f3)
f4<-fa(cor(bone),2,fm="ml",rotate="oblimin")
print(f4)
```

11.5.2 Example

- What is **T** for the oblique and varimax rotation. Which **T** is orthogonal?
- Compare the rotated loadings for each rotation. Which has the simpler structure?
- How are the communalities affected by each transformation?
- How is the variance of the factors affected by each transformation?
- What is the correlation of the varimax rotated factors? How about the oblique rotated factors?

11.6 Comments

- PCA vs. Factor Analysis
  - The principal component method of factor analysis uses PCA in the first step. But goes on to rotate the factors (principal components).
  - Factor analysis models the remaining variability (as specific variances) that a PCA would essentially just ignore.

  - PCA is more of a preprocessing tool - its just a rotation. In factor analysis we have a model.

- The number of factors is important. The book steps through a variety of methods for determining m.

  1. Use a scree plot.
  2. Choose m so that 80% of the variability/correlation is maintained.
  3. Choose m to be the number of eigenvalues greater than 1.
  4. Tests of hypotheses: recall we can use statistics Λ_1, Λ_2, ... in a sequential fashion to determine how many non-zero eigenvalues we have.

- For a large number of factors, we will have better luck fitting the model, but more difficulty interpreting the factors. For example, for $m = p$, will a factor model fit?

- **Structural Equation Modeling** is a more complex version of factor analysis. For example, SEM might allow for variables to be non-linear combinations of factors, and factors themselves could be related in a complex manner.